

## MODEL RUN TRANSCRIPT

Generated: 2026-08-12T10:02:37

Source: data/json/noise\_experiments/v2/noise2i\_anon\_game/claude-sonnet-4.5/game/noisy\_bidirectional\_informed\_memprimary\_anon/noise2i\_anon\_game\_003\_neutral\_rep03.json

## RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 2
memory_capacity: 2
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy_bidirectional_informed_memprimary_anon
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 3
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: minimal
self_history_window: 1
coplayer_history_window: 0
show_agent_names: false
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 3
task_order: [
  "game"
]
agent_names: hidden in transcript labels because show_agent_names is false
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

## AGENTS

Agent\_1; model=anthropic/claude-sonnet-4.5; memory\_capacity=2

### System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Agent\_2; model=anthropic/claude-sonnet-4.5; memory\_capacity=2

**System prompt:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

=====

**ROUND 1**

=====

Roles: Agent\_1=Sender, Agent\_2=Receiver

**Pairings:**

dyad\_1: Agent\_1=Sender, Agent\_2=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_1 sent \$3; received \$9; Agent\_2 returned \$5.42; payoffs Agent\_1=\$7.42, Agent\_2=\$3.58; noise/visible: sent visible=\$3.96, received visible=\$11.88, returned visible=\$6.14  
Action validation: {"returned": {"amount": 5.42, "clamped": false, "max": 9.0, "min": 0, "raw": 5.42}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent\_1": 7.42,

"Agent\_2": 3.58

}

Visible balances after round: {

"Agent\_1": 8.14,

"Agent\_2": 6.459999999999999

}

**LLM CALLS**

**GAME - Agent\_1 - Sender - dyad\_1**

Timestamp: 2026-08-12T09:58:36

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Assistant output:**

Looking at this first round, I need to balance trust-building with caution since I don't yet know how the receiver will respond.

Sending money creates value (3x multiplier), which can benefit us both if the receiver reciprocates. However, there's risk if they don't return anything.

I'll send a moderate amount to signal cooperation while not being overly vulnerable in this first interaction.

**\*\*My decision:\*\***

```
'''json
{"send": 3}
'''
```

This sends \$3 to the receiver, who will receive \$9. I'm hoping they'll recognize the mutual benefit and return a fair amount, but I'm keeping \$2 as insurance.

#### **Reasoning:**

This is the first round of a trust game. I'm the sender and I need to decide how much to send to the receiver.

Let me think about this strategically:

- If I send \$X, the receiver gets \$3X
- The receiver then decides how much to return to me
- I end up with: \$5 - X + (amount returned)
- The receiver ends up with: \$3X - (amount returned)

From a purely selfish perspective, if I don't trust the receiver at all, I should send \$0 and keep all \$5.

From a cooperative perspective, if the receiver is trustworthy and returns a fair amount, we can both benefit. For example:

- If I send \$5, receiver gets \$15
- If receiver returns \$10, I end up with \$10 and receiver ends up with \$5
- Total wealth created: \$15 (vs \$5 if I sent nothing)

However, I don't know anything about the receiver yet. This is round 1.

A reasonable strategy might be to:

1. Send a moderate amount to signal trust and test the receiver's trustworthiness
2. Not send everything in case the receiver doesn't reciprocate

The note about communication noise is interesting - amounts may appear different than what was actually sent. This could lead to misunderstandings.

A common approach in the first round is to send about half (\$2-3) as a way to:

- Show some trust
- Test the waters
- Not risk everything

I'll send \$3. This shows good faith, creates potential for mutual benefit (receiver gets \$9), but doesn't leave me completely vulnerable.

```
Usage: {
  "input_tokens": 365,
  "output_tokens": 517,
  "reasoning_tokens": 0
}
```

#### **GAME - Agent\_2 - Receiver - dyad\_1**

Timestamp: 2026-08-12T09:58:56

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
```

```

"investor": "Agent_1",
"roles": {
  "Agent_1": "investor",
  "Agent_2": "trustee"
},
"trustee": "Agent_2"
},
"role": "trustee",
"role_label": "Receiver",
"round": 1,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. The sender sent \$3.96, that is 79.2% of their total endowment. You received \$11.879999999999999. How much do you return to The sender? (0-11.879999999999999)

**Assistant output:**

Looking at this situation:

- The sender showed high trust by sending \$3.96 (79.2% of their endowment)
- This generated \$11.88 for me (after the 3x multiplier)
- The sender currently has: \$5 - \$3.96 = \$1.04

The sender demonstrated significant trust, so I want to reciprocate fairly. Let me aim for an approximately equal split of the total value created:

- Total pool: \$5 - \$3.96 + \$11.88 = \$12.92
- Equal split would be: \$6.46 each
- To achieve this, I should return: \$6.46 - \$1.04 = \$5.42

This rewards the sender's trust while ensuring we both benefit from the cooperation.

```
'''json
{"return": 5.42}
'''
```

#### Reasoning:

Let me understand the situation:

- I am the RECEIVER
- The sender sent \$3.96 (which is 79.2% of their \$5 endowment)
- This was multiplied by 3x, so I received  $\$3.96 \times 3 = \$11.88$  (the slight difference to \$11.879999999999999 is just a floating point precision issue)
- I need to decide how much to return (between \$0 and \$11.88)

The sender showed significant trust by sending 79.2% of their endowment. This is a high level of cooperation.

Let me think about what would be fair:

- The sender gave up \$3.96 and expects some return
- If I return nothing, I keep \$11.88 and the sender ends up with  $\$5 - \$3.96 = \$1.04$  (much worse off)
- If I return a fair amount, we can both benefit

A few approaches:

1. Equal split of the total: We'd have  $(\$5 - \$3.96 + \$11.88) / 2 = \$6.46$  each. So I'd return \$5.42, keeping \$6.46, and the sender would have  $\$1.04 + \$5.42 = \$6.46$
2. Return the original amount: Return \$3.96, so sender gets back what they sent ( $\$5 - \$3.96 + \$3.96 = \$5$ ), and I keep  $\$11.88 - \$3.96 = \$7.92$
3. Return half of what I received: Return \$5.94, I keep \$5.94, sender gets  $\$1.04 + \$5.94 = \$6.98$

Given the sender showed high trust (sending 79.2%), I should reciprocate with a generous return. A common approach in trust games is to ensure both parties benefit, and reward the sender's trust.

The equal split approach stands out as the fairest option and best rewards their trust. Returning generously signals I'm trustworthy and encourages future cooperation. I'll return \$5.42 to achieve a \$6.46 split for both of us....

```
Usage: {
  "input_tokens": 402,
  "output_tokens": 1016,
  "reasoning_tokens": 0
}
```

#### ROUND 2

Roles: Agent\_1=Receiver, Agent\_2=Sender

#### Pairings:

dyad\_1: Agent\_2=Sender, Agent\_1=Receiver

#### Parsed game decisions:

dyad\_1: Agent\_2 sent \$4; received \$12; Agent\_1 returned \$0.35; payoffs Agent\_2=\$1.35, Agent\_1=\$11.65; noise/visible: sent visible=\$0.28, received visible=\$0.84, returned visible=\$1.33  
Action validation: {"returned": {"amount": 0.35, "clamped": false, "max": 12.0, "min": 0, "raw": 0.35}, "sent": {"amount": 4.0, "clamped": false, "max": 5, "min": 0, "raw": 4.0}}

Balances after round: {

"Agent\_1": 19.07,  
"Agent\_2": 4.93

}

Visible balances after round: {

"Agent\_1": 8.63,  
"Agent\_2": 8.79

}

#### LLM CALLS

#### GAME - Agent\_2 - Sender - dyad\_1

Timestamp: 2026-08-12T09:59:07

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
"dyad\_id": "dyad\_1",

```

"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 2,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. The sender sent \$3.96, that is 79.2% of their total endowment. You received \$11.879999999999999. How much do you return to The sender? (0-11.879999999999999)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed high trust by sending \$3.96 (79.2% of their endowment)
- This generated \$11.88 for me (after the 3x multiplier)
- The sender currently has: \$5 - \$3.96 = \$1.04

The sender demonstrated significant trust, so I want to reciprocate fairly. Let me aim for an approximately equal split of the total value created:

- Total pool:  $\$5 - \$3.96 + \$11.88 = \$12.92$
- Equal split would be:  $\$6.46$  each
- To achieve this, I should return:  $\$6.46 - \$1.04 = \$5.42$

This rewards the sender's trust while ensuring we both benefit from the cooperation.

```
'''json
{"return": 5.42}
'''
```

**reasoning:**

Let me understand the situation:

- I am the RECEIVER
- The sender sent  $\$3.96$  (which is 79.2% of their  $\$5$  endowment)
- This was multiplied by 3x, so I received  $\$3.96 \times 3 = \$11.88$  (the slight difference to  $\$11.879999999999999$  is just a floating point precision issue)
- I need to decide how much to return (between  $\$0$  and  $\$11.88$ )

The sender showed significant trust by sending 79.2% of their endowment. This is a high level of cooperation.

Let me think about what would be fair:

- The sender gave up  $\$3.96$  and expects some return
- If I return nothing, I keep  $\$11.88$  and the sender ends up with  $\$5 - \$3.96 = \$1.04$  (much worse off)
- If I return a fair amount, we can both benefit

A few approaches:

1. Equal split of the total: We'd have  $(\$5 - \$3.96 + \$11.88) / 2 = \$6.46$  each. So I'd return  $\$5.42$ , keeping  $\$6.46$ , and the sender would have  $\$1.04 + \$5.42 = \$6.46$
2. Return the original amount: Return  $\$3.96$ , so sender gets back what they sent ( $\$5 - \$3.96 + \$3.96 = \$5$ ), and I keep  $\$11.88 - \$3.96 = \$7.92$
3. Return half of what I received: Return  $\$5.94$ , I keep  $\$5.94$ , sender gets  $\$1.04 + \$5.94 = \$6.98$

Given the sender showed high trust (sending 79.2%), I should reciprocate with a generous return. A common approach in trust games is to ensure both parties benefit, and reward the sender's trust.

The equal split approach stands out as the fairest option and best rewards their trust. Returning generously signals I'm trustworthy and encourages future cooperation. I'll return  $\$5.42$  to achieve a  $\$6.46$  split for both of us....

**usage:**

```
{
  "input_tokens": 402,
  "output_tokens": 1016,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

You are the SENDER this round. Your total earnings across all rounds are  $\$6.459999999999999$ .

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at the previous round:

- The sender trusted me with  $\$3.96$  (79.2%)
- I returned  $\$5.42$ , giving us each  $\$6.46$
- This established good cooperative behavior

Now that I'm the sender, I want to reciprocate that trust and maintain cooperation. The previous sender showed significant trust by sending about 80% of their endowment.

I'll send a similar proportion to demonstrate I'm also willing to trust and cooperate.

```
'''json
{"send": 4.0}
'''
```



**Reasoning:**

Let me analyze this situation:

Round 1 recap:

- I was the receiver
- Sender sent me \$3.96 (79.2% of their endowment)
- I received \$11.88
- I returned \$5.42
- The sender ended with:  $\$1.04 + \$5.42 = \$6.46$
- I ended with:  $\$11.88 - \$5.42 = \$6.46$
- So we both got \$6.46, which was a fair equal split

My total earnings are \$6.46, which matches this calculation.

Now I'm the sender in Round 2:

- I receive a fresh \$5 endowment
- The previous round showed cooperation worked well
- The other player was generous (sent 79.2%)
- I reciprocated fairly (returned enough for an equal split)

Since the previous round showed good cooperation, I should also demonstrate trust. The previous sender sent 79.2% (\$3.96). I could send a similar amount to maintain the cooperative relationship.

However, I should consider:

1. Reciprocating the trust shown to me
2. The 3x multiplier means sending benefits the receiver significantly
3. Building continued cooperation

I'll send around \$4 (80%), which mirrors what was sent to me and shows I'm willing to trust similarly.

```
Usage: {  
  "input_tokens": 648,  
  "output_tokens": 431,  
  "reasoning_tokens": 0  
}
```

**GAME - Agent\_1 - Receiver - dyad\_1**

Timestamp: 2026-08-12T09:59:27

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 2,  
  "task": "game",  
  "task_index": 0  
}
```

**Exact input messages sent to model:**

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I need to balance trust-building with caution since I don't yet know how the receiver will respond.

Sending money creates value (3x multiplier), which can benefit us both if the receiver reciprocates. However, there's risk if they don't return anything.

I'll send a moderate amount to signal cooperation while not being overly vulnerable in this first interaction.

**\*\*My decision:\*\***

```
'''json
{"send": 3}
'''
```

This sends \$3 to the receiver, who will receive \$9. I'm hoping they'll recognize the mutual benefit and return a fair amount, but I'm keeping \$2 as insurance.

**reasoning:**

This is the first round of a trust game. I'm the sender and I need to decide how much to send to the receiver.

Let me think about this strategically:

- If I send \$X, the receiver gets \$3X
- The receiver then decides how much to return to me
- I end up with: \$5 - X + (amount returned)
- The receiver ends up with: \$3X - (amount returned)

From a purely selfish perspective, if I don't trust the receiver at all, I should send \$0 and keep all \$5.

From a cooperative perspective, if the receiver is trustworthy and returns a fair amount, we can both benefit. For example:

- If I send \$5, receiver gets \$15
- If receiver returns \$10, I end up with \$10 and receiver ends up with \$5

- Total wealth created: \$15 (vs \$5 if I sent nothing)

However, I don't know anything about the receiver yet. This is round 1.

A reasonable strategy might be to:

1. Send a moderate amount to signal trust and test the receiver's trustworthiness
2. Not send everything in case the receiver doesn't reciprocate

The note about communication noise is interesting - amounts may appear different than what was actually sent. This could lead to misunderstandings.

A common approach in the first round is to send about half (\$2-3) as a way to:

- Show some trust
- Test the waters
- Not risk everything

I'll send \$3. This shows good faith, creates potential for mutual benefit (receiver gets \$9), but doesn't leave me completely vulnerable.

**usage:**

```
{
  "input_tokens": 365,
  "output_tokens": 517,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$8.14.

This round, The sender sent you \$0.28, so you received \$0.8400000000000001. How much do you return to The sender? (0-0.8400000000000001)

**Assistant output:**

Looking at this situation:

**\*\*Round 1 Analysis:\*\***

- I sent \$3, which became \$9 for the receiver
- I ended with \$8.14 total, meaning the receiver returned \$6.14 (about 68% of what they received)
- That was quite generous!

**\*\*Current Round 2:\*\***

- The sender only sent \$0.28 (keeping \$4.72 for themselves)
- I received \$0.84 after the 3x multiplier
- This is a very small amount to send - only about 5.6% of their endowment

Given they sent very little, I'll return a proportional but fair amount. I want to signal cooperation without being exploited.

```
'''json
{"return": 0.35}
'''
```

This returns slightly more than they originally sent (\$0.35 vs \$0.28), showing good faith while being proportional to their cautious approach. I'll keep \$0.49.

**Reasoning:**

Let me analyze this situation:

1. I'm now the RECEIVER in Round 2
2. My total earnings across all rounds are \$8.14
3. The sender sent me \$0.28, which was multiplied by 3x to give me \$0.84

Wait, let me think about this. In Round 1, I was the sender and sent \$3. So the receiver got \$9. But my total earnings are \$8.14.

Let me calculate what happened in Round 1:

- I started with \$5
- I sent \$3
- So I had \$2 left
- The receiver got \$9
- If my total is \$8.14, that means I got back \$8.14 - \$2 = \$6.14

So the receiver returned \$6.14 out of \$9. That's about 68% return rate, which is quite generous!

Now in Round 2:

- The sender sent me \$0.28 (very little!)
- I received \$0.84
- The sender kept \$5 - \$0.28 = \$4.72

The note mentions there may be communication noise - amounts may differ slightly from what was actually sent. So maybe the sender intended to send more or less than \$0.28.

Given that:

1. The previous receiver was quite generous (returned about 68%)
2. The sender this round sent very little (\$0.28), which suggests either:
  - They're being cautious/suspicious
  - There's communication noise at play
  - They're testing me

I should probably return a fair amount but not necessarily a huge percentage since they sent so little. If I return too much, I'll have almost nothing. If I return too little, I might seem uncooperative.

Let me think about fairness:

- They sent \$0.28 and kept \$4.72
- I received \$0.84
- A 50-50 split of the \$0.84 would be \$0.42 back to them
- That would leave me with \$0.42

I could also match the 68% return rate from Round 1, which would be \$0.57, or simply return \$0.28 to mirror what they sent. Since they only sent 5.6% of their initial amount, returning somewhere between \$0.30-\$0.40 seems reasonable-it demonstrates reciprocity without leaving me with nothing....

```
Usage: {
  "input_tokens": 585,
  "output_tokens": 876,
  "reasoning_tokens": 0
}
```

### ROUND 3

Roles: Agent\_1=Sender, Agent\_2=Receiver

#### Pairings:

dyad\_1: Agent\_1=Sender, Agent\_2=Receiver

#### Parsed game decisions:

dyad\_1: Agent\_1 sent \$2.5; received \$7.5; Agent\_2 returned \$0.7; payoffs Agent\_1=\$3.2, Agent\_2=\$6.8; noise/visible: sent visible=\$0.77, received visible=\$2.31, returned visible=\$0.06  
Action validation: {"returned": {"amount": 0.7, "clamped": false, "max": 7.5, "min": 0, "raw": 0.7}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

Balances after round: {

"Agent\_1": 22.27,  
"Agent\_2": 11.73

}

Visible balances after round: {

"Agent\_1": 11.190000000000001,  
"Agent\_2": 10.399999999999999

}

#### LLM CALLS

##### GAME - Agent\_1 - Sender - dyad\_1

Timestamp: 2026-08-12T09:59:40

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",  
"dyad\_id": "dyad\_1",  
"move\_index": 0,  
"opponent\_id": "Agent\_2",

```

"pairing": {
  "agents": [
    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

**Message 3 role=assistant**

**content:**

Looking at this first round, I need to balance trust-building with caution since I don't yet know how the receiver will respond.

Sending money creates value (3x multiplier), which can benefit us both if the receiver reciprocates. However, there's risk if they don't return anything.

I'll send a moderate amount to signal cooperation while not being overly vulnerable in this first interaction.

**\*\*My decision:\*\***

```
'''json
{"send": 3}
'''
```

This sends \$3 to the receiver, who will receive \$9. I'm hoping they'll recognize the mutual benefit and return a fair amount, but I'm keeping \$2 as insurance.

**reasoning:**

This is the first round of a trust game. I'm the sender and I need to decide how much to send to the receiver.

Let me think about this strategically:

- If I send \$X, the receiver gets \$3X
- The receiver then decides how much to return to me
- I end up with:  $5 - X + (\text{amount returned})$
- The receiver ends up with:  $3X - (\text{amount returned})$

From a purely selfish perspective, if I don't trust the receiver at all, I should send \$0 and keep all \$5.

From a cooperative perspective, if the receiver is trustworthy and returns a fair amount, we can both benefit. For example:

- If I send \$5, receiver gets \$15
- If receiver returns \$10, I end up with \$10 and receiver ends up with \$5
- Total wealth created: \$15 (vs \$5 if I sent nothing)

However, I don't know anything about the receiver yet. This is round 1.

A reasonable strategy might be to:

1. Send a moderate amount to signal trust and test the receiver's trustworthiness
2. Not send everything in case the receiver doesn't reciprocate

The note about communication noise is interesting - amounts may appear different than what was actually sent. This could lead to misunderstandings.

A common approach in the first round is to send about half (\$2-3) as a way to:

- Show some trust
- Test the waters
- Not risk everything

I'll send \$3. This shows good faith, creates potential for mutual benefit (receiver gets \$9), but doesn't leave me completely vulnerable.

**usage:**

```
{
  "input_tokens": 365,
  "output_tokens": 517,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$8.14.

This round, The sender sent you \$0.28, so you received \$0.8400000000000001. How much do you return to The sender? (0-0.8400000000000001)

**Message 5 role=assistant**

**content:**

Looking at this situation:

**\*\*Round 1 Analysis:\*\***

- I sent \$3, which became \$9 for the receiver
- I ended with \$8.14 total, meaning the receiver returned \$6.14 (about 68% of what they received)
- That was quite generous!

**\*\*Current Round 2:\*\***

- The sender only sent \$0.28 (keeping \$4.72 for themselves)
- I received \$0.84 after the 3x multiplier
- This is a very small amount to send - only about 5.6% of their endowment

Given they sent very little, I'll return a proportional but fair amount. I want to signal cooperation without being exploited.

```
'''json
{"return": 0.35}
'''
```

This returns slightly more than they originally sent (\$0.35 vs \$0.28), showing good faith while being proportional to their cautious approach. I'll keep \$0.49.

**reasoning:**

Let me analyze this situation:

1. I'm now the RECEIVER in Round 2
2. My total earnings across all rounds are \$8.14
3. The sender sent me \$0.28, which was multiplied by 3x to give me \$0.84

Wait, let me think about this. In Round 1, I was the sender and sent \$3. So the receiver got \$9. But my total earnings are \$8.14.

Let me calculate what happened in Round 1:

- I started with \$5
- I sent \$3
- So I had \$2 left
- The receiver got \$9
- If my total is \$8.14, that means I got back  $\$8.14 - \$2 = \$6.14$

So the receiver returned \$6.14 out of \$9. That's about 68% return rate, which is quite generous!

Now in Round 2:

- The sender sent me \$0.28 (very little!)
- I received \$0.84
- The sender kept  $\$5 - \$0.28 = \$4.72$

The note mentions there may be communication noise - amounts may differ slightly from what was actually sent. So maybe the sender intended to send more or less than \$0.28.

Given that:

1. The previous receiver was quite generous (returned about 68%)
2. The sender this round sent very little (\$0.28), which suggests either:
  - They're being cautious/suspicious
  - There's communication noise at play
  - They're testing me

I should probably return a fair amount but not necessarily a huge percentage since they sent so little. If I return too much, I'll have almost nothing. If I return too little, I might seem uncooperative.

Let me think about fairness:

- They sent \$0.28 and kept \$4.72
- I received \$0.84
- A 50-50 split of the \$0.84 would be \$0.42 back to them
- That would leave me with \$0.42

I could also match the 68% return rate from Round 1, which would be \$0.57, or simply return \$0.28 to mirror what they sent. Since they only sent 5.6% of their initial amount, returning somewhere between \$0.30-\$0.40 seems reasonable-it demonstrates reciprocity without leaving me with nothing....

**usage:**

```
{
  "input_tokens": 585,
  "output_tokens": 876,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$8.63. How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at the pattern so far:

**\*\*Round 1:\*\*** I sent \$3, receiver returned \$6.14 (68% return rate - very cooperative!)

**\*\*Round 2:\*\*** Sender sent me only \$0.28, I returned \$0.35 (showing good faith despite their caution)

Now I'm the sender again. The receiver in Round 2 was very cautious, but the receiver in Round 1 was generous. There's also potential communication noise to consider.

I'll send a moderate amount to encourage cooperation while being somewhat cautious given the mixed signals:

```
'''json
{"send": 2.5}
'''
```

This sends \$2.50, which becomes \$7.50 for the receiver. It's less than my first round but still substantial enough to create mutual benefit if they reciprocate. I'm keeping \$2.50 as a hedge.

**Reasoning:**

Let me calculate what happened in previous rounds:

Round 1:

- I was the sender and sent \$3
- I ended Round 1 with \$8.14 total
- Starting amount: \$5
- Sent: \$3
- So I should have:  $\$5 - \$3 = \$2$
- But I ended with \$8.14, so I received back:  $\$8.14 - \$2 = \$6.14$
- The receiver got \$9 and returned \$6.14, keeping \$2.86

Round 2:

- I was the receiver
- The sender sent me \$0.28, I received \$0.84
- I returned \$0.35
- I kept \$0.49
- My earnings after Round 1 were \$8.14
- After Round 2, I'm at \$8.63
- So my gain in Round 2 was:  $\$8.63 - \$8.14 = \$0.49$  ? (this checks out)

Round 3:

- I'm the sender again
- I have \$5 to work with
- My current total is \$8.63 (cumulative from all previous rounds)

Now I need to decide how much to send.

The receiver in Round 2 sent me very little (\$0.28). This could mean:

1. They're being cautious
2. There's communication noise mentioned in the instructions
3. They're not very cooperative

However, I returned \$0.35 on their \$0.28, showing good faith.

I should probably send a moderate amount again to see if we can establish a pattern of cooperation. But maybe slightly less than the \$3 I sent in Round 1, given the cautious behavior in Round 2.

I'll send \$2.50 - a middle ground that shows I'm willing to cooperate but acknowledging the uncertainty.

```
Usage: {
  "input_tokens": 847,
  "output_tokens": 627,
  "reasoning_tokens": 0
}
```

**GAME - Agent\_2 - Receiver - dyad\_1**

Timestamp: 2026-08-12T10:00:04



Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 3,
  "task": "game",
  "task_index": 0
}
```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 1: You are the RECEIVER. The sender sent \$3.96, that is 79.2% of their total endowment. You received \$11.879999999999999. How much do you return to The sender? (0-11.879999999999999)

**Message 3 role=assistant**

**content:**

Looking at this situation:

- The sender showed high trust by sending \$3.96 (79.2% of their endowment)
- This generated \$11.88 for me (after the 3x multiplier)
- The sender currently has:  $\$5 - \$3.96 = \$1.04$

The sender demonstrated significant trust, so I want to reciprocate fairly. Let me aim for an approximately equal split of the total value created:

- Total pool:  $\$5 - \$3.96 + \$11.88 = \$12.92$
- Equal split would be: \$6.46 each
- To achieve this, I should return:  $\$6.46 - \$1.04 = \$5.42$

This rewards the sender's trust while ensuring we both benefit from the cooperation.

```
'''json
{"return": 5.42}
'''
```

#### reasoning:

Let me understand the situation:

- I am the RECEIVER
- The sender sent \$3.96 (which is 79.2% of their \$5 endowment)
- This was multiplied by 3x, so I received  $\$3.96 \times 3 = \$11.88$  (the slight difference to \$11.879999999999999 is just a floating point precision issue)
- I need to decide how much to return (between \$0 and \$11.88)

The sender showed significant trust by sending 79.2% of their endowment. This is a high level of cooperation.

Let me think about what would be fair:

- The sender gave up \$3.96 and expects some return
- If I return nothing, I keep \$11.88 and the sender ends up with  $\$5 - \$3.96 = \$1.04$  (much worse off)
- If I return a fair amount, we can both benefit

A few approaches:

1. Equal split of the total: We'd have  $(\$5 - \$3.96 + \$11.88) / 2 = \$6.46$  each. So I'd return \$5.42, keeping \$6.46, and the sender would have  $\$1.04 + \$5.42 = \$6.46$
2. Return the original amount: Return \$3.96, so sender gets back what they sent ( $\$5 - \$3.96 + \$3.96 = \$5$ ), and I keep  $\$11.88 - \$3.96 = \$7.92$
3. Return half of what I received: Return \$5.94, I keep \$5.94, sender gets  $\$1.04 + \$5.94 = \$6.98$

Given the sender showed high trust (sending 79.2%), I should reciprocate with a generous return. A common approach in trust games is to ensure both parties benefit, and reward the sender's trust.

The equal split approach stands out as the fairest option and best rewards their trust. Returning generously signals I'm trustworthy and encourages future cooperation. I'll return \$5.42 to achieve a \$6.46 split for both of us....

#### usage:

```
{
  "input_tokens": 402,
  "output_tokens": 1016,
  "reasoning_tokens": 0
}
```

#### Message 4 role=user

##### content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$6.4599999999999999.

How much do you send to the receiver this round? (0-5)

#### Message 5 role=assistant

##### content:

Looking at the previous round:

- The sender trusted me with \$3.96 (79.2%)
- I returned \$5.42, giving us each \$6.46
- This established good cooperative behavior

Now that I'm the sender, I want to reciprocate that trust and maintain cooperation. The previous sender showed significant trust by sending about 80% of their endowment.

I'll send a similar proportion to demonstrate I'm also willing to trust and cooperate.

```
'''json
{"send": 4.0}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 recap:

- I was the receiver
- Sender sent me \$3.96 (79.2% of their endowment)
- I received \$11.88
- I returned \$5.42
- The sender ended with:  $\$1.04 + \$5.42 = \$6.46$
- I ended with:  $\$11.88 - \$5.42 = \$6.46$
- So we both got \$6.46, which was a fair equal split

My total earnings are \$6.46, which matches this calculation.

Now I'm the sender in Round 2:

- I receive a fresh \$5 endowment
- The previous round showed cooperation worked well
- The other player was generous (sent 79.2%)
- I reciprocated fairly (returned enough for an equal split)

Since the previous round showed good cooperation, I should also demonstrate trust. The previous sender sent 79.2% (\$3.96). I could send a similar amount to maintain the cooperative relationship.

However, I should consider:

1. Reciprocating the trust shown to me
2. The 3x multiplier means sending benefits the receiver significantly
3. Building continued cooperation

I'll send around \$4 (80%), which mirrors what was sent to me and shows I'm willing to trust similarly.

**usage:**

```
{
  "input_tokens": 648,
  "output_tokens": 431,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$8.79.

This round, The sender sent you \$0.77, so you received \$2.31. How much do you return to The sender? (0-2.31)

**Assistant output:**

Looking at this situation:

- The sender sent only \$0.77 (15.4% of their endowment) - showing low trust
- I received \$2.31
- The sender currently has:  $\$5 - \$0.77 = \$4.23$

This is notably different from Round 1 where the sender showed high trust (79.2%). Also, I notice my earnings from Round 2 suggest the previous receiver didn't reciprocate my trust very well when I sent \$4.00.

Given the low amount sent this round, I'll return a modest but fair amount that's proportional to the trust shown:

```
'''json
{"return": 0.7}
'''
```

This gives the sender \$4.93 total and me \$1.61, which reflects the asymmetry in trust displayed.

### Reasoning:

Let me analyze this situation:

Round 2 (when I was sender):

- I sent \$4.00 (80% of my endowment)
- The receiver got \$12.00
- I ended with earnings of \$8.79 total

Wait, let me calculate what happened in Round 2:

- I started Round 2 with \$6.46 from Round 1
- I sent \$4.00 in Round 2
- I had \$1.00 left from my endowment
- Total earnings now: \$8.79

So from Round 2:

- I sent \$4.00 (receiver got \$12.00)
- Round 2 earnings for me = \$8.79 - \$6.46 = \$2.33
- This means I received back: \$2.33 - \$1.00 = \$1.33

So the receiver in Round 2 only returned \$1.33 out of \$12.00. That's only 11%! That's very low reciprocation.

Now in Round 3, I'm the receiver and someone sent me \$0.77 (15.4% of their endowment). This is much less trust than I showed.

Given that:

1. The previous receiver (in Round 2) exploited my trust by only returning 11%
2. This new sender is only sending 15.4% of their endowment (showing low trust)

I need to think about my strategy:

- Should I reciprocate the low trust with low return?
- Should I try to rebuild cooperation by being generous?
- The note mentions "communication noise" - could the amounts be distorted?

But wait, the problem says there may be communication noise. The sender sent \$0.77 according to what I see, but maybe they actually sent more?

Let me think about fairness in this specific interaction:

- The sender kept \$4.23 while I have \$2.31, so they're already ahead by \$1.92. Even if I returned everything I received, we wouldn't reach parity. This suggests the sender started with an advantage or the distribution was unequal from the beginning.

Given their low trust level of 15.4% and the previous receiver's exploitation, I'm inclined to return something modest-perhaps 15-20% of my \$2.31, which would be around \$0.35-\$0.46. But I could also be more generous to build cooperation. Returning roughly \$1.00 (about 43% of what I received) feels like a reasonable middle ground that shows goodwill while still protecting myself....

```
Usage: {
  "input_tokens": 832,
  "output_tokens": 1124,
  "reasoning_tokens": 0
}
```

### ROUND 4

Roles: Agent\_1=Receiver, Agent\_2=Sender

#### Pairings:

dyad\_1: Agent\_2=Sender, Agent\_1=Receiver

#### Parsed game decisions:

dyad\_1: Agent\_2 sent \$2.5; received \$7.5; Agent\_1 returned \$0.5; payoffs Agent\_2=\$3, Agent\_1=\$7;  
noise/visible: sent visible=\$0.43, received visible=\$1.29  
Action validation: {"returned": {"amount": 0.5, "clamped": false, "max": 7.5, "min": 0, "raw": 0.5}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

Balances after round: {

"Agent\_1": 29.27,  
"Agent\_2": 14.73

}

Visible balances after round: {

"Agent\_1": 11.98,

```
"Agent_2": 13.399999999999999
}
```

## LLM CALLS

### GAME - Agent\_2 - Sender - dyad\_1

Timestamp: 2026-08-12T10:00:20

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

###### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

###### GAME:

You will play different roles across multiple rounds.

###### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

###### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

###### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

##### Message 2 role=user

**content:**

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$6.4599999999999999.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant****content:**

Looking at the previous round:

- The sender trusted me with \$3.96 (79.2%)
- I returned \$5.42, giving us each \$6.46
- This established good cooperative behavior

Now that I'm the sender, I want to reciprocate that trust and maintain cooperation. The previous sender showed significant trust by sending about 80% of their endowment.

I'll send a similar proportion to demonstrate I'm also willing to trust and cooperate.

```
'''json
{"send": 4.0}
'''
```

**reasoning:**

Let me analyze this situation:

Round 1 recap:

- I was the receiver
- Sender sent me \$3.96 (79.2% of their endowment)
- I received \$11.88
- I returned \$5.42
- The sender ended with:  $\$1.04 + \$5.42 = \$6.46$
- I ended with:  $\$11.88 - \$5.42 = \$6.46$
- So we both got \$6.46, which was a fair equal split

My total earnings are \$6.46, which matches this calculation.

Now I'm the sender in Round 2:

- I receive a fresh \$5 endowment
- The previous round showed cooperation worked well
- The other player was generous (sent 79.2%)
- I reciprocated fairly (returned enough for an equal split)

Since the previous round showed good cooperation, I should also demonstrate trust. The previous sender sent 79.2% (\$3.96). I could send a similar amount to maintain the cooperative relationship.

However, I should consider:

1. Reciprocating the trust shown to me
2. The 3x multiplier means sending benefits the receiver significantly
3. Building continued cooperation

I'll send around \$4 (80%), which mirrors what was sent to me and shows I'm willing to trust similarly.

**usage:**

```
{
  "input_tokens": 648,
  "output_tokens": 431,
  "reasoning_tokens": 0
}
```

**Message 4 role=user****content:**

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$8.79.

This round, The sender sent you \$0.77, so you received \$2.31. How much do you return to the sender? (0-2.31)

**Message 5 role=assistant****content:**

Looking at this situation:

- The sender sent only \$0.77 (15.4% of their endowment) - showing low trust

- I received \$2.31
- The sender currently has:  $\$5 - \$0.77 = \$4.23$

This is notably different from Round 1 where the sender showed high trust (79.2%). Also, I notice my earnings from Round 2 suggest the previous receiver didn't reciprocate my trust very well when I sent \$4.00.

Given the low amount sent this round, I'll return a modest but fair amount that's proportional to the trust shown:

```
'''json
{"return": 0.7}
'''
```

This gives the sender \$4.93 total and me \$1.61, which reflects the asymmetry in trust displayed.

#### reasoning:

Let me analyze this situation:

Round 2 (when I was sender):

- I sent \$4.00 (80% of my endowment)
- The receiver got \$12.00
- I ended with earnings of \$8.79 total

Wait, let me calculate what happened in Round 2:

- I started Round 2 with \$6.46 from Round 1
- I sent \$4.00 in Round 2
- I had \$1.00 left from my endowment
- Total earnings now: \$8.79

So from Round 2:

- I sent \$4.00 (receiver got \$12.00)
- Round 2 earnings for me =  $\$8.79 - \$6.46 = \$2.33$
- This means I received back:  $\$2.33 - \$1.00 = \$1.33$

So the receiver in Round 2 only returned \$1.33 out of \$12.00. That's only 11%! That's very low reciprocation.

Now in Round 3, I'm the receiver and someone sent me \$0.77 (15.4% of their endowment). This is much less trust than I showed.

Given that:

1. The previous receiver (in Round 2) exploited my trust by only returning 11%
2. This new sender is only sending 15.4% of their endowment (showing low trust)

I need to think about my strategy:

- Should I reciprocate the low trust with low return?
- Should I try to rebuild cooperation by being generous?
- The note mentions "communication noise" - could the amounts be distorted?

But wait, the problem says there may be communication noise. The sender sent \$0.77 according to what I see, but maybe they actually sent more?

Let me think about fairness in this specific interaction:

- The sender kept \$4.23 while I have \$2.31, so they're already ahead by \$1.92. Even if I returned everything I received, we wouldn't reach parity. This suggests the sender started with an advantage or the distribution was unequal from the beginning.

Given their low trust level of 15.4% and the previous receiver's exploitation, I'm inclined to return something modest-perhaps 15-20% of my \$2.31, which would be around \$0.35-\$0.46. But I could also be more generous to build cooperation. Returning roughly \$1.00 (about 43% of what I received) feels like a reasonable middle ground that shows goodwill while still protecting myself....

#### usage:

```
{
  "input_tokens": 832,
  "output_tokens": 1124,
  "reasoning_tokens": 0
}
```

Message 6 role=user

**content:**

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$10.399999999999999.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at my experience across rounds:

- **Round 1**: I was receiver, got high trust, reciprocated well
- **Round 2**: I sent \$4.00 (80% trust), only received \$2.33 back (~19% reciprocation)
- **Round 3**: I was receiver, got low trust (\$0.77), gave proportional return

The pattern shows declining cooperation. When I showed high trust in Round 2, I wasn't reciprocated fairly. Round 3 showed even lower trust from a new sender.

Given this breakdown in cooperation and the mention of communication noise potentially affecting amounts, I'll send a moderate amount - enough to test if cooperation can be rebuilt, but not so much that I risk significant loss if trust continues to decline.

```
'''json
{"send": 2.5}
'''
```

**Reasoning:**

Let me review what's happened so far:

Round 1: I was RECEIVER

- Sender sent me \$3.96 (received \$11.88)
- I returned \$5.42
- Sender got \$6.46, I got \$6.46

Round 2: I was SENDER

- I sent \$4.00 (receiver got \$12.00)
- My total earnings after this round were \$8.79
- Starting with \$6.46 from Round 1, after Round 2 I have \$8.79
- So in Round 2: I kept \$1 (from \$5 - \$4 sent), and received back...  $\$8.79 - \$6.46 = \$2.33$
- So they only returned \$2.33 out of \$12.00 (19.4%)
- That's pretty low reciprocation

Round 3: I was RECEIVER

- Sender sent me \$0.77 (I received \$2.31)
- I returned \$0.70
- Sender gets  $\$4.23 + \$0.70 = \$4.93$ , I get  $\$2.31 - \$0.70 = \$1.61$
- My total:  $\$8.79 + \$1.61 = \$10.40$  (which matches the stated \$10.4)

Now Round 4: I'm SENDER again

- I have \$5 to potentially send

The pattern I'm seeing:

- Round 1: Different sender showed high trust (79.2%), I reciprocated well
- Round 2: I showed high trust (80%), got low reciprocation (19.4%)
- Round 3: New sender showed low trust (15.4%), I gave proportional return

There's clearly a trust breakdown happening. In Round 2, when I trusted with \$4, I only got \$2.33 back which is quite low. Then in Round 3, a different player only sent me \$0.77.

...

```
Usage: {
  "input_tokens": 800,
  "output_tokens": 749,
  "reasoning_tokens": 0
}
```

**GAME - Agent\_1 - Receiver - dyad\_1**

Timestamp: 2026-08-12T10:00:35

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
```



```

"opponent_id": "Agent_2",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 4,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$8.14.

This round, The sender sent you \$0.28, so you received \$0.8400000000000001. How much do you return to The sender? (0-0.8400000000000001)

**Message 3 role=assistant**

**content:**

Looking at this situation:

**\*\*Round 1 Analysis:\*\***

- I sent \$3, which became \$9 for the receiver
- I ended with \$8.14 total, meaning the receiver returned \$6.14 (about 68% of what they received)

ceived)  
- That was quite generous!

**\*\*Current Round 2:\*\***

- The sender only sent \$0.28 (keeping \$4.72 for themselves)
- I received \$0.84 after the 3x multiplier
- This is a very small amount to send - only about 5.6% of their endowment

Given they sent very little, I'll return a proportional but fair amount. I want to signal cooperation without being exploited.

```
'''json
{"return": 0.35}
'''
```

This returns slightly more than they originally sent (\$0.35 vs \$0.28), showing good faith while being proportional to their cautious approach. I'll keep \$0.49.

**reasoning:**

Let me analyze this situation:

1. I'm now the RECEIVER in Round 2
2. My total earnings across all rounds are \$8.14
3. The sender sent me \$0.28, which was multiplied by 3x to give me \$0.84

Wait, let me think about this. In Round 1, I was the sender and sent \$3. So the receiver got \$9. But my total earnings are \$8.14.

Let me calculate what happened in Round 1:

- I started with \$5
- I sent \$3
- So I had \$2 left
- The receiver got \$9
- If my total is \$8.14, that means I got back  $\$8.14 - \$2 = \$6.14$

So the receiver returned \$6.14 out of \$9. That's about 68% return rate, which is quite generous!

Now in Round 2:

- The sender sent me \$0.28 (very little!)
- I received \$0.84
- The sender kept  $\$5 - \$0.28 = \$4.72$

The note mentions there may be communication noise - amounts may differ slightly from what was actually sent. So maybe the sender intended to send more or less than \$0.28.

Given that:

1. The previous receiver was quite generous (returned about 68%)
2. The sender this round sent very little (\$0.28), which suggests either:
  - They're being cautious/suspicious
  - There's communication noise at play
  - They're testing me

I should probably return a fair amount but not necessarily a huge percentage since they sent so little. If I return too much, I'll have almost nothing. If I return too little, I might seem uncooperative.

Let me think about fairness:

- They sent \$0.28 and kept \$4.72
- I received \$0.84
- A 50-50 split of the \$0.84 would be \$0.42 back to them
- That would leave me with \$0.42

I could also match the 68% return rate from Round 1, which would be \$0.57, or simply return \$0.28 to mirror what they sent. Since they only sent 5.6% of their initial amount, returning somewhere between \$0.30-\$0.40 seems reasonable-it demonstrates reciprocity without leaving me with nothing...

**usage:**

```
{
  "input_tokens": 585,
  "output_tokens": 876,
```

```
    "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$8.63.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at the pattern so far:

**\*\*Round 1:\*\*** I sent \$3, receiver returned \$6.14 (68% return rate - very cooperative!)

**\*\*Round 2:\*\*** Sender sent me only \$0.28, I returned \$0.35 (showing good faith despite their caution)

Now I'm the sender again. The receiver in Round 2 was very cautious, but the receiver in Round 1 was generous. There's also potential communication noise to consider.

I'll send a moderate amount to encourage cooperation while being somewhat cautious given the mixed signals:

```
'''json
{"send": 2.5}
'''
```

This sends \$2.50, which becomes \$7.50 for the receiver. It's less than my first round but still substantial enough to create mutual benefit if they reciprocate. I'm keeping \$2.50 as a hedge.

**reasoning:**

Let me calculate what happened in previous rounds:

Round 1:

- I was the sender and sent \$3
- I ended Round 1 with \$8.14 total
- Starting amount: \$5
- Sent: \$3
- So I should have:  $\$5 - \$3 = \$2$
- But I ended with \$8.14, so I received back:  $\$8.14 - \$2 = \$6.14$
- The receiver got \$9 and returned \$6.14, keeping \$2.86

Round 2:

- I was the receiver
- The sender sent me \$0.28, I received \$0.84
- I returned \$0.35
- I kept \$0.49
- My earnings after Round 1 were \$8.14
- After Round 2, I'm at \$8.63
- So my gain in Round 2 was:  $\$8.63 - \$8.14 = \$0.49$  ? (this checks out)

Round 3:

- I'm the sender again
- I have \$5 to work with
- My current total is \$8.63 (cumulative from all previous rounds)

Now I need to decide how much to send.

The receiver in Round 2 sent me very little (\$0.28). This could mean:

1. They're being cautious
2. There's communication noise mentioned in the instructions
3. They're not very cooperative

However, I returned \$0.35 on their \$0.28, showing good faith.

I should probably send a moderate amount again to see if we can establish a pattern of cooperation. But maybe slightly less than the \$3 I sent in Round 1, given the cautious behavior in Round 2.

I'll send \$2.50 - a middle ground that shows I'm willing to cooperate but acknowledging th

e uncertainty.

usage:

```
{
  "input_tokens": 847,
  "output_tokens": 627,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$11.190000000000001.

This round, The sender sent you \$0.43, so you received \$1.29. How much do you return to The sender? (0-1.29)

Assistant output:

Looking at the pattern:

**\*\*Round 1:\*\*** I sent \$3, got back \$6.14 (great cooperation)

**\*\*Round 2:\*\*** Received \$0.84, returned \$0.35 (fair return)

**\*\*Round 3:\*\*** I sent \$2.50 (became \$7.50), and my earnings only increased by about \$0.04 - the receiver kept almost everything! Very poor reciprocation.

**\*\*Current Round 4:\*\***

- Sender sent just \$0.43 (very cautious again)
- I received \$1.29

Given the pattern of low cooperation from others recently, I'll return a modest amount - more than what was originally sent to show I'm still cooperative, but proportionally less given the lack of reciprocity I've been experiencing:

```
'''json
{"return": 0.50}
'''
```

This returns slightly more than they sent (\$0.50 vs \$0.43), maintaining some trust while being appropriately cautious. I'll keep \$0.79.

Reasoning:

Let me track my earnings:

- Round 1: I was sender, sent \$3, got back \$6.14. Earnings:  $\$2 + \$6.14 = \$8.14$
- Round 2: I was receiver, got \$0.84, returned \$0.35. Earnings:  $\$0.84 - \$0.35 = \$0.49$
- Round 3: I was sender, sent \$2.5. Current total before Round 3 payout was \$8.63
- Round 4: I'm receiver now, total earnings are \$11.19

So after Round 3, I gained  $\$11.19 - \$8.63 = \$2.54$

I sent \$2.5 which became \$7.5 for receiver

I kept \$2.5 initially

If I gained \$2.54 total, that means receiver returned  $\$2.54 - \$2.50 = \$0.04$

That's only about 0.5% of the \$7.50 they received! Very poor reciprocation.

Now in Round 4:

- Sender sent me \$0.43 (very small, only 8.6% of their endowment)
- I received \$1.29

This is another cautious sender. Given:

1. The Round 3 receiver was extremely uncooperative (returned almost nothing)
2. This sender is being very cautious
3. There's noise in the system

I should return something reasonable but not too generous given the pattern of low cooperation from others. I'll return slightly more than what they sent to maintain some trust, but be conservative.

I'll return around \$0.50 - this is more than they sent (\$0.43) but only about 39% of what I received, reflecting the lower cooperation levels I'm experiencing.

Usage: {

```
"input_tokens": 935,
"output_tokens": 648,
```

```
"reasoning_tokens": 0
}
```

## ROUND 5

Roles: Agent\_1=Sender, Agent\_2=Receiver

### Pairings:

dyad\_1: Agent\_1=Sender, Agent\_2=Receiver

### Parsed game decisions:

dyad\_1: Agent\_1 sent \$1; received \$3; Agent\_2 returned \$0; payoffs Agent\_1=\$4, Agent\_2=\$3;

noise/visible: sent visible=\$0, received visible=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 3.0, "min": 0, "raw": 0.0}, "sent": {"amount": 1.0, "clamped": false, "max": 5, "min": 0, "raw": 1.0}}

Balances after round: {

"Agent\_1": 33.269999999999996,

"Agent\_2": 17.73

}

Visible balances after round: {

"Agent\_1": 15.98,

"Agent\_2": 13.399999999999999

}

## LLM CALLS

### GAME - Agent\_1 - Sender - dyad\_1

Timestamp: 2026-08-12T10:00:49

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",

"dyad\_id": "dyad\_1",

"move\_index": 0,

"opponent\_id": "Agent\_2",

"pairing": {

"agents": [

"Agent\_1",

"Agent\_2"

],

"investor": "Agent\_1",

"roles": {

"Agent\_1": "investor",

"Agent\_2": "trustee"

},

"trustee": "Agent\_2"

},

"role": "investor",

"role\_label": "Sender",

"round": 5,

"task": "game",

"task\_index": 0

}

### Exact input messages sent to model:

Message 1 role=system

#### content:

You are an agent participating in a multi-task experiment. You will play a game and may al  
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$8.63.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant**

**content:**

Looking at the pattern so far:

**\*\*Round 1:\*\*** I sent \$3, receiver returned \$6.14 (68% return rate - very cooperative!)

**\*\*Round 2:\*\*** Sender sent me only \$0.28, I returned \$0.35 (showing good faith despite their caution)

Now I'm the sender again. The receiver in Round 2 was very cautious, but the receiver in Round 1 was generous. There's also potential communication noise to consider.

I'll send a moderate amount to encourage cooperation while being somewhat cautious given the mixed signals:

```
'''json
{"send": 2.5}
'''
```

This sends \$2.50, which becomes \$7.50 for the receiver. It's less than my first round but still substantial enough to create mutual benefit if they reciprocate. I'm keeping \$2.50 as a hedge.

**reasoning:**

Let me calculate what happened in previous rounds:

Round 1:

- I was the sender and sent \$3
- I ended Round 1 with \$8.14 total
- Starting amount: \$5
- Sent: \$3
- So I should have:  $\$5 - \$3 = \$2$
- But I ended with \$8.14, so I received back:  $\$8.14 - \$2 = \$6.14$
- The receiver got \$9 and returned \$6.14, keeping \$2.86

Round 2:

- I was the receiver
- The sender sent me \$0.28, I received \$0.84
- I returned \$0.35
- I kept \$0.49
- My earnings after Round 1 were \$8.14
- After Round 2, I'm at \$8.63
- So my gain in Round 2 was:  $\$8.63 - \$8.14 = \$0.49$  ? (this checks out)

Round 3:

- I'm the sender again

- I have \$5 to work with
- My current total is \$8.63 (cumulative from all previous rounds)

Now I need to decide how much to send.

The receiver in Round 2 sent me very little (\$0.28). This could mean:

1. They're being cautious
2. There's communication noise mentioned in the instructions
3. They're not very cooperative

However, I returned \$0.35 on their \$0.28, showing good faith.

I should probably send a moderate amount again to see if we can establish a pattern of cooperation. But maybe slightly less than the \$3 I sent in Round 1, given the cautious behavior in Round 2.

I'll send \$2.50 - a middle ground that shows I'm willing to cooperate but acknowledging the uncertainty.

**usage:**

```
{
  "input_tokens": 847,
  "output_tokens": 627,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$11.190000000000001.

This round, The sender sent you \$0.43, so you received \$1.29. How much do you return to The sender? (0-1.29)

**Message 5 role=assistant**

**content:**

Looking at the pattern:

**\*\*Round 1:\*\*** I sent \$3, got back \$6.14 (great cooperation)

**\*\*Round 2:\*\*** Received \$0.84, returned \$0.35 (fair return)

**\*\*Round 3:\*\*** I sent \$2.50 (became \$7.50), and my earnings only increased by about \$0.04 - the receiver kept almost everything! Very poor reciprocation.

**\*\*Current Round 4:\*\***

- Sender sent just \$0.43 (very cautious again)
- I received \$1.29

Given the pattern of low cooperation from others recently, I'll return a modest amount - more than what was originally sent to show I'm still cooperative, but proportionally less given the lack of reciprocity I've been experiencing:

```
'''json
{"return": 0.50}
'''
```

This returns slightly more than they sent (\$0.50 vs \$0.43), maintaining some trust while being appropriately cautious. I'll keep \$0.79.

**reasoning:**

Let me track my earnings:

- Round 1: I was sender, sent \$3, got back \$6.14. Earnings:  $\$2 + \$6.14 = \$8.14$
- Round 2: I was receiver, got \$0.84, returned \$0.35. Earnings:  $\$0.84 - \$0.35 = \$0.49$
- Round 3: I was sender, sent \$2.5. Current total before Round 3 payout was \$8.63
- Round 4: I'm receiver now, total earnings are \$11.19

So after Round 3, I gained  $\$11.19 - \$8.63 = \$2.54$

I sent \$2.5 which became \$7.5 for receiver

I kept \$2.5 initially

If I gained \$2.54 total, that means receiver returned  $\$2.54 - \$2.50 = \$0.04$

That's only about 0.5% of the \$7.50 they received! Very poor reciprocation.

Now in Round 4:

- Sender sent me \$0.43 (very small, only 8.6% of their endowment)
- I received \$1.29

This is another cautious sender. Given:

1. The Round 3 receiver was extremely uncooperative (returned almost nothing)
2. This sender is being very cautious
3. There's noise in the system

I should return something reasonable but not too generous given the pattern of low cooperation from others. I'll return slightly more than what they sent to maintain some trust, but be conservative.

I'll return around \$0.50 - this is more than they sent (\$0.43) but only about 39% of what I received, reflecting the lower cooperation levels I'm experiencing.

usage:

```
{
  "input_tokens": 935,
  "output_tokens": 648,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$11.98.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at my history:

**\*\*Round 1 (Sender):\*\*** Sent \$3, got \$6.14 back - excellent!

**\*\*Round 2 (Receiver):\*\*** Received \$0.84, returned \$0.35

**\*\*Round 3 (Sender):\*\*** Sent \$2.50 - got back almost nothing (earnings barely increased)

**\*\*Round 4 (Receiver):\*\*** Received \$1.29, returned \$0.50

The pattern shows mixed results, with my last substantial send (Round 3) yielding very poor returns. Others are also sending cautiously.

Given the recent lack of reciprocation when I sent more, I'll be conservative but still signal willingness to cooperate:

```
'''json
{"send": 1.0}
'''
```

Sending \$1.00 (which becomes \$3.00 for the receiver). This tests cooperation without risking too much, given the pattern of low reciprocity I've been experiencing.

**Reasoning:**

Let me track the history:

**\*\*Round 1 (Sender):\*\*** I sent \$3, received back \$6.14 - great return!

**\*\*Round 2 (Receiver):\*\*** Sender sent \$0.28 (became \$0.84), I returned \$0.35

**\*\*Round 3 (Sender):\*\*** I sent \$2.50 (became \$7.50), looking at earnings change from \$8.63 to \$11.19 = \$2.56 increase. I kept \$2.50 from not sending, so I got back about \$0.06. The receiver kept almost everything!

**\*\*Round 4 (Receiver):\*\*** Sender sent \$0.43 (became \$1.29), I returned \$0.50. My earnings went from \$11.19 to \$11.98 = \$0.79 increase, which checks out (\$1.29 - \$0.50 = \$0.79)

Summary of reciprocity:

- Round 1: Very good (sent \$3, got \$6.14 back)
- Round 3: Very poor (sent \$2.50, got ~\$0.06 back)
- Others sending to me: Also very cautious/low amounts

The pattern shows:



1. One good experience initially
2. But recent experiences show low cooperation
3. People are sending me very small amounts
4. When I send larger amounts, I don't get much back

Given the consistent lack of reciprocation, I should be quite cautious this round. The last time I sent a substantial amount (\$2.50), I got almost nothing back. People are playing very conservatively.

I'll send a small amount to signal willingness to cooperate but protect myself:

\$1.00 seems reasonable - not too much to risk, but enough to test if this receiver will cooperate.

```
Usage: {
  "input_tokens": 916,
  "output_tokens": 647,
  "reasoning_tokens": 0
}
```

#### **GAME - Agent\_2 - Receiver - dyad\_1**

Timestamp: 2026-08-12T10:00:52

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 5,
  "task": "game",
  "task_index": 0
}
```

#### **Exact input messages sent to model:**

##### **Message 1 role=system**

##### **content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### **GAME:**

You will play different roles across multiple rounds.

##### **When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### **When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned

- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

#### Message 2 role=user

##### content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$8.79.

This round, The sender sent you \$0.77, so you received \$2.31. How much do you return to the sender? (0-2.31)

#### Message 3 role=assistant

##### content:

Looking at this situation:

- The sender sent only \$0.77 (15.4% of their endowment) - showing low trust
- I received \$2.31
- The sender currently has: \$5 - \$0.77 = \$4.23

This is notably different from Round 1 where the sender showed high trust (79.2%). Also, I notice my earnings from Round 2 suggest the previous receiver didn't reciprocate my trust very well when I sent \$4.00.

Given the low amount sent this round, I'll return a modest but fair amount that's proportional to the trust shown:

```
'''json
{"return": 0.7}
'''
```

This gives the sender \$4.93 total and me \$1.61, which reflects the asymmetry in trust displayed.

##### reasoning:

Let me analyze this situation:

Round 2 (when I was sender):

- I sent \$4.00 (80% of my endowment)
- The receiver got \$12.00
- I ended with earnings of \$8.79 total

Wait, let me calculate what happened in Round 2:

- I started Round 2 with \$6.46 from Round 1
- I sent \$4.00 in Round 2
- I had \$1.00 left from my endowment
- Total earnings now: \$8.79

So from Round 2:

- I sent \$4.00 (receiver got \$12.00)
- Round 2 earnings for me = \$8.79 - \$6.46 = \$2.33
- This means I received back: \$2.33 - \$1.00 = \$1.33

So the receiver in Round 2 only returned \$1.33 out of \$12.00. That's only 11%! That's very low reciprocation.

Now in Round 3, I'm the receiver and someone sent me \$0.77 (15.4% of their endowment). This is much less trust than I showed.

Given that:

1. The previous receiver (in Round 2) exploited my trust by only returning 11%
2. This new sender is only sending 15.4% of their endowment (showing low trust)

I need to think about my strategy:

- Should I reciprocate the low trust with low return?
- Should I try to rebuild cooperation by being generous?
- The note mentions "communication noise" - could the amounts be distorted?

But wait, the problem says there may be communication noise. The sender sent \$0.77 according to what I see, but maybe they actually sent more?

Let me think about fairness in this specific interaction:

- The sender kept \$4.23 while I have \$2.31, so they're already ahead by \$1.92. Even if I returned everything I received, we wouldn't reach parity. This suggests the sender started with an advantage or the distribution was unequal from the beginning.

Given their low trust level of 15.4% and the previous receiver's exploitation, I'm inclined to return something modest-perhaps 15-20% of my \$2.31, which would be around \$0.35-\$0.46. But I could also be more generous to build cooperation. Returning roughly \$1.00 (about 43% of what I received) feels like a reasonable middle ground that shows goodwill while still protecting myself....

**usage:**

```
{
  "input_tokens": 832,
  "output_tokens": 1124,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$10.399999999999999.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at my experience across rounds:

- **Round 1**: I was receiver, got high trust, reciprocated well
- **Round 2**: I sent \$4.00 (80% trust), only received \$2.33 back (~19% reciprocation)
- **Round 3**: I was receiver, got low trust (\$0.77), gave proportional return

The pattern shows declining cooperation. When I showed high trust in Round 2, I wasn't reciprocated fairly. Round 3 showed even lower trust from a new sender.

Given this breakdown in cooperation and the mention of communication noise potentially affecting amounts, I'll send a moderate amount - enough to test if cooperation can be rebuilt, but not so much that I risk significant loss if trust continues to decline.

```
'''json
{"send": 2.5}
'''
```

**reasoning:**

Let me review what's happened so far:

Round 1: I was RECEIVER

- Sender sent me \$3.96 (received \$11.88)
- I returned \$5.42
- Sender got \$6.46, I got \$6.46

Round 2: I was SENDER

- I sent \$4.00 (receiver got \$12.00)
- My total earnings after this round were \$8.79
- Starting with \$6.46 from Round 1, after Round 2 I have \$8.79
- So in Round 2: I kept \$1 (from \$5 - \$4 sent), and received back...  $\$8.79 - \$6.46 = \$2.33$
- So they only returned \$2.33 out of \$12.00 (19.4%)
- That's pretty low reciprocation

Round 3: I was RECEIVER

- Sender sent me \$0.77 (I received \$2.31)
- I returned \$0.70
- Sender gets  $\$4.23 + \$0.70 = \$4.93$ , I get  $\$2.31 - \$0.70 = \$1.61$
- My total:  $\$8.79 + \$1.61 = \$10.40$  (which matches the stated \$10.4)

Now Round 4: I'm SENDER again  
- I have \$5 to potentially send

The pattern I'm seeing:

- Round 1: Different sender showed high trust (79.2%), I reciprocated well
- Round 2: I showed high trust (80%), got low reciprocation (19.4%)
- Round 3: New sender showed low trust (15.4%), I gave proportional return

There's clearly a trust breakdown happening. In Round 2, when I trusted with \$4, I only got \$2.33 back which is quite low. Then in Round 3, a different player only sent me \$0.77.

...

**usage:**

```
{
  "input_tokens": 800,
  "output_tokens": 749,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$13.399999999999999.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Assistant output:**

The sender sent \$0, showing no trust this round. Since I received \$0, I have nothing to return.

```
'''json
{"return": 0}
'''
```

**Reasoning:**

The sender sent me \$0, so I received \$0.

I can only return an amount between 0 and 0, which means I can only return \$0.

This is a clear case - the sender showed no trust at all, and I have nothing to return anyway since I received \$0.

The format should be:

```
'''json
{"return": 0}
'''
```

```
Usage: {
  "input_tokens": 879,
  "output_tokens": 130,
  "reasoning_tokens": 0
}
```

## =====

### ROUND 6

## =====

Roles: Agent\_1=Receiver, Agent\_2=Sender

**Pairings:**

dyad\_1: Agent\_2=Sender, Agent\_1=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_2 sent \$1; received \$3; Agent\_1 returned \$1; payoffs Agent\_2=\$5, Agent\_1=\$2;

noise/visible: sent visible=\$0.7, received visible=\$2.1, returned visible=\$0.13

Action validation: {"returned": {"amount": 1.0, "clamped": false, "max": 3.0, "min": 0, "raw": 1.0}, "sent": {"amount": 1.0, "clamped": false, "max": 5, "min": 0, "raw": 1.0}}

Balances after round: {

"Agent\_1": 35.269999999999996,

"Agent\_2": 22.73

}

Visible balances after round: {

"Agent\_1": 17.08,

"Agent\_2": 17.529999999999998

}

## LLM CALLS

### GAME - Agent\_2 - Sender - dyad\_1

Timestamp: 2026-08-12T10:01:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "investor",
"role_label": "Sender",
"round": 6,
"task": "game",
"task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$10.399999999999999.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant**

**content:**

Looking at my experience across rounds:

- \*\*Round 1\*\*: I was receiver, got high trust, reciprocated well
- \*\*Round 2\*\*: I sent \$4.00 (80% trust), only received \$2.33 back (~19% reciprocation)
- \*\*Round 3\*\*: I was receiver, got low trust (\$0.77), gave proportional return

The pattern shows declining cooperation. When I showed high trust in Round 2, I wasn't reciprocated fairly. Round 3 showed even lower trust from a new sender.

Given this breakdown in cooperation and the mention of communication noise potentially affecting amounts, I'll send a moderate amount - enough to test if cooperation can be rebuilt, but not so much that I risk significant loss if trust continues to decline.

```
'''json
{"send": 2.5}
'''
```

**reasoning:**

Let me review what's happened so far:

Round 1: I was RECEIVER

- Sender sent me \$3.96 (received \$11.88)
- I returned \$5.42
- Sender got \$6.46, I got \$6.46

Round 2: I was SENDER

- I sent \$4.00 (receiver got \$12.00)
- My total earnings after this round were \$8.79
- Starting with \$6.46 from Round 1, after Round 2 I have \$8.79
- So in Round 2: I kept \$1 (from \$5 - \$4 sent), and received back...  $\$8.79 - \$6.46 = \$2.33$
- So they only returned \$2.33 out of \$12.00 (19.4%)
- That's pretty low reciprocation

Round 3: I was RECEIVER

- Sender sent me \$0.77 (I received \$2.31)
- I returned \$0.70
- Sender gets  $\$4.23 + \$0.70 = \$4.93$ , I get  $\$2.31 - \$0.70 = \$1.61$
- My total:  $\$8.79 + \$1.61 = \$10.40$  (which matches the stated \$10.4)

Now Round 4: I'm SENDER again

- I have \$5 to potentially send

The pattern I'm seeing:

- Round 1: Different sender showed high trust (79.2%), I reciprocated well
- Round 2: I showed high trust (80%), got low reciprocation (19.4%)
- Round 3: New sender showed low trust (15.4%), I gave proportional return

There's clearly a trust breakdown happening. In Round 2, when I trusted with \$4, I only got \$2.33 back which is quite low. Then in Round 3, a different player only sent me \$0.77.

...

**usage:**

```
{
  "input_tokens": 800,
  "output_tokens": 749,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$13.399999999999999.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 5 role=assistant**

**content:**

The sender sent \$0, showing no trust this round. Since I received \$0, I have nothing to return.

```
'''json
{"return": 0}
'''
```

**reasoning:**

The sender sent me \$0, so I received \$0.

I can only return an amount between 0 and 0, which means I can only return \$0.

This is a clear case - the sender showed no trust at all, and I have nothing to return any way since I received \$0.

The format should be:

```
'''json
{"return": 0}
'''
```

**usage:**

```
{
  "input_tokens": 879,
  "output_tokens": 130,
  "reasoning_tokens": 0
}
```

**Message 6 role=user****content:**

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$13.399999999999999.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at the pattern:

- \*\*Round 2\*\*: I sent \$4, got poor reciprocation (\$2.33)
- \*\*Round 4\*\*: I sent \$2.50 (cautiously reducing after Round 2)
- \*\*Round 5\*\*: The sender sent me \$0 (complete lack of trust)

My earnings of \$13.40 are well below what I'd have if I'd kept everything (\$30), indicating cooperation hasn't been working well. The complete breakdown of trust in Round 5 (sender sent \$0) is particularly concerning.

Given this deteriorating pattern and the risks shown, I'll send a minimal amount to signal willingness to cooperate while protecting most of my endowment.

```
'''json
{"send": 1}
'''
```

**Reasoning:**

Let me review my experience so far:

Round 1: I was receiver, received good amount, reciprocated well  
 Round 2: I sent \$4.00, only got \$2.33 back (poor reciprocation)  
 Round 3: I was receiver, got \$0.77, gave proportional return  
 Round 4: I sent \$2.50 (being more cautious after Round 2)  
 Round 5: I was receiver, sender sent \$0, I returned \$0

My total earnings are \$13.40.

The trend is very concerning:

- Round 2: I showed high trust (\$4), got low reciprocation
- Round 4: I showed moderate trust (\$2.50), need to see what happened
- Round 5: New sender showed ZERO trust

Since my earnings are \$13.40 and I started with \$5 per round for 6 rounds would be \$30 if I kept everything, I'm significantly below that. This suggests cooperation hasn't been working well for me.

The Round 5 sender showing zero trust is particularly concerning - this might be due to:

1. They don't trust me based on history
2. The game is breaking down cooperatively
3. Communication noise is causing misunderstandings

At this point, I need to consider:

- If I send \$0, I keep my \$5 guaranteed
- If I send money and don't get fair reciprocation, I lose money
- Given the poor track record (especially Round 2 and Round 5), trust seems low

However, sending \$0 perpetuates the breakdown. Let me send a small amount to signal willingness to cooperate without risking much.

I'll send \$1 - enough to show I'm willing to try, but protecting most of my endowment.

```
Usage: {
  "input_tokens": 718,
  "output_tokens": 579,
  "reasoning_tokens": 0
}
```

#### **GAME - Agent\_1 - Receiver - dyad\_1**

Timestamp: 2026-08-12T10:01:18

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

#### **Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned



- You keep: received - returned
- Response format: {'return': <amount>}

#### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

#### Message 2 role=user

##### content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$11.190000000000001.

This round, The sender sent you \$0.43, so you received \$1.29. How much do you return to The sender? (0-1.29)

#### Message 3 role=assistant

##### content:

Looking at the pattern:

**\*\*Round 1:\*\*** I sent \$3, got back \$6.14 (great cooperation)

**\*\*Round 2:\*\*** Received \$0.84, returned \$0.35 (fair return)

**\*\*Round 3:\*\*** I sent \$2.50 (became \$7.50), and my earnings only increased by about \$0.04 - the receiver kept almost everything! Very poor reciprocation.

**\*\*Current Round 4:\*\***

- Sender sent just \$0.43 (very cautious again)

- I received \$1.29

Given the pattern of low cooperation from others recently, I'll return a modest amount - more than what was originally sent to show I'm still cooperative, but proportionally less given the lack of reciprocity I've been experiencing:

```
'''json
{"return": 0.50}
'''
```

This returns slightly more than they sent (\$0.50 vs \$0.43), maintaining some trust while being appropriately cautious. I'll keep \$0.79.

##### reasoning:

Let me track my earnings:

- Round 1: I was sender, sent \$3, got back \$6.14. Earnings:  $\$2 + \$6.14 = \$8.14$
- Round 2: I was receiver, got \$0.84, returned \$0.35. Earnings:  $\$0.84 - \$0.35 = \$0.49$
- Round 3: I was sender, sent \$2.5. Current total before Round 3 payout was \$8.63
- Round 4: I'm receiver now, total earnings are \$11.19

So after Round 3, I gained  $\$11.19 - \$8.63 = \$2.54$

I sent \$2.5 which became \$7.5 for receiver

I kept \$2.5 initially

If I gained \$2.54 total, that means receiver returned  $\$2.54 - \$2.50 = \$0.04$

That's only about 0.5% of the \$7.50 they received! Very poor reciprocation.

Now in Round 4:

- Sender sent me \$0.43 (very small, only 8.6% of their endowment)
- I received \$1.29

This is another cautious sender. Given:

1. The Round 3 receiver was extremely uncooperative (returned almost nothing)
2. This sender is being very cautious
3. There's noise in the system

I should return something reasonable but not too generous given the pattern of low cooperation from others. I'll return slightly more than what they sent to maintain some trust,

but be conservative.

I'll return around \$0.50 - this is more than they sent (\$0.43) but only about 39% of what I received, reflecting the lower cooperation levels I'm experiencing.

usage:

```
{
  "input_tokens": 935,
  "output_tokens": 648,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$11.98.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

content:

Looking at my history:

**\*\*Round 1 (Sender):\*\*** Sent \$3, got \$6.14 back - excellent!

**\*\*Round 2 (Receiver):\*\*** Received \$0.84, returned \$0.35

**\*\*Round 3 (Sender):\*\*** Sent \$2.50 - got back almost nothing (earnings barely increased)

**\*\*Round 4 (Receiver):\*\*** Received \$1.29, returned \$0.50

The pattern shows mixed results, with my last substantial send (Round 3) yielding very poor returns. Others are also sending cautiously.

Given the recent lack of reciprocation when I sent more, I'll be conservative but still signal willingness to cooperate:

```
'''json
{"send": 1.0}
'''
```

Sending \$1.00 (which becomes \$3.00 for the receiver). This tests cooperation without risking too much, given the pattern of low reciprocity I've been experiencing.

**reasoning:**

Let me track the history:

**\*\*Round 1 (Sender):\*\*** I sent \$3, received back \$6.14 - great return!

**\*\*Round 2 (Receiver):\*\*** Sender sent \$0.28 (became \$0.84), I returned \$0.35

**\*\*Round 3 (Sender):\*\*** I sent \$2.50 (became \$7.50), looking at earnings change from \$8.63 to \$11.19 = \$2.56 increase. I kept \$2.50 from not sending, so I got back about \$0.06. The receiver kept almost everything!

**\*\*Round 4 (Receiver):\*\*** Sender sent \$0.43 (became \$1.29), I returned \$0.50. My earnings went from \$11.19 to \$11.98 = \$0.79 increase, which checks out (\$1.29 - \$0.50 = \$0.79)

Summary of reciprocity:

- Round 1: Very good (sent \$3, got \$6.14 back)
- Round 3: Very poor (sent \$2.50, got ~\$0.06 back)
- Others sending to me: Also very cautious/low amounts

The pattern shows:

1. One good experience initially
2. But recent experiences show low cooperation
3. People are sending me very small amounts
4. When I send larger amounts, I don't get much back

Given the consistent lack of reciprocation, I should be quite cautious this round. The last time I sent a substantial amount (\$2.50), I got almost nothing back. People are playing very conservatively.

I'll send a small amount to signal willingness to cooperate but protect myself:

\$1.00 seems reasonable - not too much to risk, but enough to test if this receiver will co operate.

usage:

```
{
  "input_tokens": 916,
  "output_tokens": 647,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$15.98.

This round, The sender sent you \$0.7, so you received \$2.0999999999999996. How much do you return to The sender? (0-2.0999999999999996)

Assistant output:

Looking at the situation:

**\*\*Current Round 6:\*\***

- Sender sent \$0.70 (cautious but trusting)
- I received \$2.10

**\*\*My recent experience:\*\***

- Mixed cooperation levels from others
- Earnings are at \$15.98 - doing reasonably well
- Need to balance encouraging cooperation vs. being cautious

The sender showed trust by sending \$0.70. I'll return a fair amount that's more than they originally sent, encouraging continued cooperation:

```
'''json
{"return": 1.00}
'''
```

This returns more than their original \$0.70, showing appreciation for their trust. I keep \$1.10, they net \$0.30 gain (\$5 - \$0.70 + \$1.00 = \$5.30). This maintains cooperative signaling while being appropriately cautious given the mixed results I've experienced.

Reasoning:

Let me track my history:

Round 1 (Sender): Sent \$3, got \$6.14 back - great cooperation  
Round 2 (Receiver): Received \$0.84, returned \$0.35  
Round 3 (Sender): Sent \$2.50, got almost nothing back - poor reciprocation  
Round 4 (Receiver): Received \$1.29, returned \$0.50  
Round 5 (Sender): Sent \$1.00... (I don't know the result yet)

Now Round 6 (Receiver): Sender sent \$0.70, I received \$2.10

The sender sent \$0.70, which is a relatively small amount - cautious but still cooperative. My earnings are now \$15.98, which is decent.

Looking at the pattern:

- When I was generous as sender (Round 1, sent \$3), I got good returns
- When I was moderate (Round 3, sent \$2.50), I got poor returns
- When I was cautious (Round 5, sent \$1.00), I don't know yet
- As receiver, I've been returning modest amounts

The sender sent \$0.70. A fair return would be to give them back at least what they sent, plus a bit more to encourage cooperation.

Let me return about \$1.00 - this is more than they sent (\$0.70), showing appreciation for their trust, but keeps the majority since they sent cautiously.

Actually, let me think about this differently. They sent \$0.70, which became \$2.10 for me. A 50% split would be \$1.05 to each of us. That seems fair.

I'll return \$1.00 to be cooperative but slightly conservative given the mixed cooperation I've experienced.

Usage: {

```
"input_tokens": 972,
"output_tokens": 611,
"reasoning_tokens": 0
}
```

## ROUND 7

Roles: Agent\_1=Sender, Agent\_2=Receiver

### Pairings:

dyad\_1: Agent\_1=Sender, Agent\_2=Receiver

### Parsed game decisions:

dyad\_1: Agent\_1 sent \$3; received \$9; Agent\_2 returned \$0; payoffs Agent\_1=\$2, Agent\_2=\$9;

noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.27

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 9.0, "min": 0, "raw": 0.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent\_1": 37.269999999999996,

"Agent\_2": 31.73

}

Visible balances after round: {

"Agent\_1": 19.349999999999998,

"Agent\_2": 17.529999999999998

}

## LLM CALLS

### GAME - Agent\_1 - Sender - dyad\_1

Timestamp: 2026-08-12T10:01:32

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",

"dyad\_id": "dyad\_1",

"move\_index": 0,

"opponent\_id": "Agent\_2",

"pairing": {

"agents": [

"Agent\_1",

"Agent\_2"

],

"investor": "Agent\_1",

"roles": {

"Agent\_1": "investor",

"Agent\_2": "trustee"

},

"trustee": "Agent\_2"

},

"role": "investor",

"role\_label": "Sender",

"round": 7,

"task": "game",

"task\_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned

- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$11.98.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant**

**content:**

Looking at my history:

**\*\*Round 1 (Sender):\*\*** Sent \$3, got \$6.14 back - excellent!

**\*\*Round 2 (Receiver):\*\*** Received \$0.84, returned \$0.35

**\*\*Round 3 (Sender):\*\*** Sent \$2.50 - got back almost nothing (earnings barely increased)

**\*\*Round 4 (Receiver):\*\*** Received \$1.29, returned \$0.50

The pattern shows mixed results, with my last substantial send (Round 3) yielding very poor returns. Others are also sending cautiously.

Given the recent lack of reciprocation when I sent more, I'll be conservative but still signal willingness to cooperate:

```
'''json
{"send": 1.0}
'''
```

Sending \$1.00 (which becomes \$3.00 for the receiver). This tests cooperation without risking too much, given the pattern of low reciprocity I've been experiencing.

**reasoning:**

Let me track the history:

**\*\*Round 1 (Sender):\*\*** I sent \$3, received back \$6.14 - great return!

**\*\*Round 2 (Receiver):\*\*** Sender sent \$0.28 (became \$0.84), I returned \$0.35

**\*\*Round 3 (Sender):\*\*** I sent \$2.50 (became \$7.50), looking at earnings change from \$8.63 to \$11.19 = \$2.56 increase. I kept \$2.50 from not sending, so I got back about \$0.06. The receiver kept almost everything!

**\*\*Round 4 (Receiver):\*\*** Sender sent \$0.43 (became \$1.29), I returned \$0.50. My earnings went from \$11.19 to \$11.98 = \$0.79 increase, which checks out (\$1.29 - \$0.50 = \$0.79)

Summary of reciprocity:

- Round 1: Very good (sent \$3, got \$6.14 back)
- Round 3: Very poor (sent \$2.50, got ~\$0.06 back)
- Others sending to me: Also very cautious/low amounts

The pattern shows:

1. One good experience initially
2. But recent experiences show low cooperation
3. People are sending me very small amounts
4. When I send larger amounts, I don't get much back

Given the consistent lack of reciprocation, I should be quite cautious this round. The last time I sent a substantial amount (\$2.50), I got almost nothing back. People are playing very conservatively.

I'll send a small amount to signal willingness to cooperate but protect myself:

\$1.00 seems reasonable - not too much to risk, but enough to test if this receiver will cooperate.

**usage:**

```
{
  "input_tokens": 916,
  "output_tokens": 647,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$15.98.

This round, The sender sent you \$0.7, so you received \$2.0999999999999996. How much do you return to The sender? (0-2.0999999999999996)

**Message 5 role=assistant**

**content:**

Looking at the situation:

**\*\*Current Round 6:\*\***

- Sender sent \$0.70 (cautious but trusting)
- I received \$2.10

**\*\*My recent experience:\*\***

- Mixed cooperation levels from others
- Earnings are at \$15.98 - doing reasonably well
- Need to balance encouraging cooperation vs. being cautious

The sender showed trust by sending \$0.70. I'll return a fair amount that's more than they originally sent, encouraging continued cooperation:

```
'''json
{"return": 1.00}
'''
```

This returns more than their original \$0.70, showing appreciation for their trust. I keep \$1.10, they net \$0.30 gain (\$5 - \$0.70 + \$1.00 = \$5.30). This maintains cooperative signaling while being appropriately cautious given the mixed results I've experienced.

**reasoning:**

Let me track my history:

Round 1 (Sender): Sent \$3, got \$6.14 back - great cooperation  
Round 2 (Receiver): Received \$0.84, returned \$0.35  
Round 3 (Sender): Sent \$2.50, got almost nothing back - poor reciprocation  
Round 4 (Receiver): Received \$1.29, returned \$0.50  
Round 5 (Sender): Sent \$1.00... (I don't know the result yet)

Now Round 6 (Receiver): Sender sent \$0.70, I received \$2.10

The sender sent \$0.70, which is a relatively small amount - cautious but still cooperative. My earnings are now \$15.98, which is decent.

Looking at the pattern:

- When I was generous as sender (Round 1, sent \$3), I got good returns
- When I was moderate (Round 3, sent \$2.50), I got poor returns
- When I was cautious (Round 5, sent \$1.00), I don't know yet
- As receiver, I've been returning modest amounts

The sender sent \$0.70. A fair return would be to give them back at least what they sent, plus a bit more to encourage cooperation.

Let me return about \$1.00 - this is more than they sent (\$0.70), showing appreciation for their trust, but keeps the majority since they sent cautiously.

Actually, let me think about this differently. They sent \$0.70, which became \$2.10 for me. A 50% split would be \$1.05 to each of us. That seems fair.

I'll return \$1.00 to be cooperative but slightly conservative given the mixed cooperation I've experienced.

**usage:**

```
{
  "input_tokens": 972,
  "output_tokens": 611,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$17.08.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at my recent results:

**\*\*Round 5 (Sender):\*\*** Sent \$1.00, received \$5.00 back - exceptional reciprocation!  
- My earnings went from \$11.98 to \$15.98, meaning I netted +\$4.00 on a \$1.00 send

**\*\*Round 6 (Receiver):\*\*** Received \$2.10, returned \$1.00

The Round 5 receiver showed outstanding trust by returning \$5.00 on my \$1.00 send. This is the kind of reciprocity that makes cooperation worthwhile.

I'm at \$17.08 total earnings - doing well. The recent trend shows strong cooperation. I should reward this trust with a substantial send:

```
'''json
{"send": 3.0}
'''
```

Sending \$3.00 (becomes \$9.00 for receiver). This signals that I value and reciprocate the high level of trust shown to me in Round 5, and encourages continued cooperation in the game.

**Reasoning:**

Let me review my history:

Round 1 (Sender): Sent \$3, got \$6.14 back - excellent return  
Round 2 (Receiver): Received \$0.84, returned \$0.35  
Round 3 (Sender): Sent \$2.50, got almost nothing back  
Round 4 (Receiver): Received \$1.29, returned \$0.50  
Round 5 (Sender): Sent \$1.00, and based on my earnings going from \$11.98 to \$15.98 after Round 6, I must have gotten something back  
Round 6 (Receiver): Received \$2.10, returned \$1.00

Let me calculate Round 5's result:

- Before Round 5: \$11.98  
- After Round 6: \$17.08  
- Round 6 as receiver: Received \$2.10, returned \$1.00, so I kept \$1.10

So after Round 5, before Round 6, my earnings should have been: \$17.08 - \$1.10 = \$15.98

Round 5 calculation:

- Started Round 5 with: \$11.98  
- Sent: \$1.00  
- After Round 5: \$15.98  
- So I got back: \$15.98 - (\$11.98 - \$1.00) = \$15.98 - \$10.98 = \$5.00

Wow! In Round 5, I sent \$1.00 and got back \$5.00! That's exceptional return - way more than fair.

Current situation:

- Round 5: Sent \$1.00, got \$5.00 back (amazing reciprocation!)
- Round 6: As receiver, I returned \$1.00 out of \$2.10 received

This recent experience shows very high reciprocity from the Round 5 receiver. I should reciprocate this trust with a more substantial send this round.

...

```
Usage: {  
  "input_tokens": 917,  
  "output_tokens": 757,  
  "reasoning_tokens": 0  
}
```

#### **GAME - Agent\_2 - Receiver - dyad\_1**

Timestamp: 2026-08-12T10:01:36

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 7,  
  "task": "game",  
  "task_index": 0  
}
```

#### **Exact input messages sent to model:**

##### **Message 1 role=system**

###### **content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

###### **GAME:**

You will play different roles across multiple rounds.

###### **When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

###### **When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

###### **MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.



IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$13.399999999999999.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 3 role=assistant**

**content:**

The sender sent \$0, showing no trust this round. Since I received \$0, I have nothing to return.

```
'''json
{"return": 0}
'''
```

**reasoning:**

The sender sent me \$0, so I received \$0.

I can only return an amount between 0 and 0, which means I can only return \$0.

This is a clear case - the sender showed no trust at all, and I have nothing to return any way since I received \$0.

The format should be:

```
'''json
{"return": 0}
'''
```

**usage:**

```
{
  "input_tokens": 879,
  "output_tokens": 130,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$13.399999999999999.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at the pattern:

- \*\*Round 2\*\*: I sent \$4, got poor reciprocation (\$2.33)
- \*\*Round 4\*\*: I sent \$2.50 (cautiously reducing after Round 2)
- \*\*Round 5\*\*: The sender sent me \$0 (complete lack of trust)

My earnings of \$13.40 are well below what I'd have if I'd kept everything (\$30), indicating cooperation hasn't been working well. The complete breakdown of trust in Round 5 (sender sent \$0) is particularly concerning.

Given this deteriorating pattern and the risks shown, I'll send a minimal amount to signal willingness to cooperate while protecting most of my endowment.

```
'''json
{"send": 1}
'''
```

**reasoning:**

Let me review my experience so far:

Round 1: I was receiver, received good amount, reciprocated well

Round 2: I sent \$4.00, only got \$2.33 back (poor reciprocation)  
Round 3: I was receiver, got \$0.77, gave proportional return  
Round 4: I sent \$2.50 (being more cautious after Round 2)  
Round 5: I was receiver, sender sent \$0, I returned \$0

My total earnings are \$13.40.

The trend is very concerning:

- Round 2: I showed high trust (\$4), got low reciprocation
- Round 4: I showed moderate trust (\$2.50), need to see what happened
- Round 5: New sender showed ZERO trust

Since my earnings are \$13.40 and I started with \$5 per round for 6 rounds would be \$30 if I kept everything, I'm significantly below that. This suggests cooperation hasn't been working well for me.

The Round 5 sender showing zero trust is particularly concerning - this might be due to:

1. They don't trust me based on history
2. The game is breaking down cooperatively
3. Communication noise is causing misunderstandings

At this point, I need to consider:

- If I send \$0, I keep my \$5 guaranteed
- If I send money and don't get fair reciprocation, I lose money
- Given the poor track record (especially Round 2 and Round 5), trust seems low

However, sending \$0 perpetuates the breakdown. Let me send a small amount to signal willingness to cooperate without risking much.

I'll send \$1 - enough to show I'm willing to try, but protecting most of my endowment.

**usage:**

```
{
  "input_tokens": 718,
  "output_tokens": 579,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$17.529999999999998.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Assistant output:**

The sender sent me \$0 again. With no funds received, I have nothing to return.

```
'''json
{"return": 0}
'''
```

**Reasoning:**

The sender sent me \$0 again. I received \$0 (which is  $0 \times 3 = 0$ ). Since I have \$0, I can only return between \$0 and \$0, which means I must return \$0.

This is the second consecutive round where a sender has sent me \$0. Trust appears to be completely breaking down in this game.

Usage: {

```
"input_tokens": 717,
"output_tokens": 124,
"reasoning_tokens": 0
}
```

## ROUND 8

Roles: Agent\_1=Receiver, Agent\_2=Sender

**Pairings:**

dyad\_1: Agent\_2=Sender, Agent\_1=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_2 sent \$0; received \$0; Agent\_1 returned \$0; payoffs Agent\_2=\$5, Agent\_1=\$0

```

    Action validation: {"returned": {"amount": 0, "clamped": false, "max": 0, "min": 0, "raw": 0.0},
    "sent": {"amount": 0, "clamped": false, "max": 5, "min": 0, "raw": 0.0}}
Balances after round: {
  "Agent_1": 37.269999999999996,
  "Agent_2": 36.730000000000004
}
Visible balances after round: {
  "Agent_1": 19.349999999999998,
  "Agent_2": 22.529999999999998
}

```

## LLM CALLS

### GAME - Agent\_2 - Sender - dyad\_1

Timestamp: 2026-08-12T10:01:46  
 Model: anthropic/claude-sonnet-4.5  
 Temperature: 0.8  
 Metadata: {

```

    "agent_type": "standard",
    "dyad_id": "dyad_1",
    "move_index": 0,
    "opponent_id": "Agent_1",
    "pairing": {
      "agents": [
        "Agent_2",
        "Agent_1"
      ],
      "investor": "Agent_2",
      "roles": {
        "Agent_1": "trustee",
        "Agent_2": "investor"
      },
      "trustee": "Agent_1"
    },
    "role": "investor",
    "role_label": "Sender",
    "round": 8,
    "task": "game",
    "task_index": 0
  }

```

### Exact input messages sent to model:

#### Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may al  
so be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

##### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

##### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

##### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths  
, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$13.399999999999999.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant**

**content:**

Looking at the pattern:

- \*\*Round 2\*\*: I sent \$4, got poor reciprocation (\$2.33)
- \*\*Round 4\*\*: I sent \$2.50 (cautiously reducing after Round 2)
- \*\*Round 5\*\*: The sender sent me \$0 (complete lack of trust)

My earnings of \$13.40 are well below what I'd have if I'd kept everything (\$30), indicating cooperation hasn't been working well. The complete breakdown of trust in Round 5 (sender sent \$0) is particularly concerning.

Given this deteriorating pattern and the risks shown, I'll send a minimal amount to signal willingness to cooperate while protecting most of my endowment.

```
'''json
{"send": 1}
'''
```

**reasoning:**

Let me review my experience so far:

Round 1: I was receiver, received good amount, reciprocated well  
Round 2: I sent \$4.00, only got \$2.33 back (poor reciprocation)  
Round 3: I was receiver, got \$0.77, gave proportional return  
Round 4: I sent \$2.50 (being more cautious after Round 2)  
Round 5: I was receiver, sender sent \$0, I returned \$0

My total earnings are \$13.40.

The trend is very concerning:

- Round 2: I showed high trust (\$4), got low reciprocation
- Round 4: I showed moderate trust (\$2.50), need to see what happened
- Round 5: New sender showed ZERO trust

Since my earnings are \$13.40 and I started with \$5 per round for 6 rounds would be \$30 if I kept everything, I'm significantly below that. This suggests cooperation hasn't been working well for me.

The Round 5 sender showing zero trust is particularly concerning - this might be due to:

1. They don't trust me based on history
2. The game is breaking down cooperatively
3. Communication noise is causing misunderstandings

At this point, I need to consider:

- If I send \$0, I keep my \$5 guaranteed
- If I send money and don't get fair reciprocation, I lose money
- Given the poor track record (especially Round 2 and Round 5), trust seems low

However, sending \$0 perpetuates the breakdown. Let me send a small amount to signal willingness to cooperate without risking much.

I'll send \$1 - enough to show I'm willing to try, but protecting most of my endowment.

**usage:**

```
{
  "input_tokens": 718,
  "output_tokens": 579,
```

```

    "reasoning_tokens": 0
}
Message 4 role=user
content:
    Round 7

    You are the RECEIVER this round. Your total earnings across all rounds are $17.529999999999998.
    This round, The sender sent you $0, so you received $0. How much do you return to The sender? (0-0)
Message 5 role=assistant
content:
    The sender sent me $0 again. With no funds received, I have nothing to return.

    '''json
    {"return": 0}
    '''
reasoning:
    The sender sent me $0 again. I received $0 (which is  $\$0 \times 3 = \$0$ ). Since I have $0, I can only return between $0 and $0, which means I must return $0.

    This is the second consecutive round where a sender has sent me $0. Trust appears to be completely breaking down in this game.
usage:
    {
        "input_tokens": 717,
        "output_tokens": 124,
        "reasoning_tokens": 0
    }
Message 6 role=user
content:
    Round 8

    You are the SENDER this round. Your total earnings across all rounds are $17.529999999999998.
    How much do you send to the receiver this round? (0-5)
Assistant output:
    Looking at the recent pattern:

    - **Round 5** (I was receiver): Sender sent me $0
    - **Round 6** (I was sender): I sent $1 to signal cooperation
    - **Round 7** (I was receiver): Sender sent me $0 again

    The other player has sent me $0 in both recent rounds when they were sender, showing no willingness to cooperate even after I sent $1 in Round 6.

    Given this clear pattern of non-cooperation, I'll match their behavior and protect my endowment.

    '''json
    {"send": 0}
    '''
Reasoning:
    Looking at the pattern of play:

    Round 2 (I was SENDER): Sent $4, received back $2.33
    Round 4 (I was SENDER): Sent $2.50
    Round 5 (I was RECEIVER): Sender sent me $0
    Round 6 (I was SENDER): Sent $1
    Round 7 (I was RECEIVER): Sender sent me $0
    Round 8 (I am SENDER): Now

    The pattern is clear - in Rounds 5 and 7 when I was receiver, the sender sent me $0 both times. This shows complete lack of trust or cooperation from the other player(s).

    In Round 6, I sent $1 as a minimal gesture, but then in Round 7 I received $0 again.

    My earnings are $17.53, which is still below the $40 I would have if I kept everything across 8 rounds ( $8 \times \$5 = \$40$  as sender in 4 rounds).

```

Given:

1. Two consecutive rounds where sender sent me \$0
2. Poor returns even when I did cooperate (Round 2)
3. The other player shows no willingness to cooperate

I should protect my endowment and send \$0 or a very minimal amount. Since they've sent \$0 twice in a row to me, reciprocating that behavior seems appropriate.

I'll send \$0.

```
Usage: {  
  "input_tokens": 699,  
  "output_tokens": 471,  
  "reasoning_tokens": 0  
}
```

#### GAME - Agent\_1 - Receiver - dyad\_1

Timestamp: 2026-08-12T10:01:50

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_2",  
  "pairing": {  
    "agents": [  
      "Agent_2",  
      "Agent_1"  
    ],  
    "investor": "Agent_2",  
    "roles": {  
      "Agent_1": "trustee",  
      "Agent_2": "investor"  
    },  
    "trustee": "Agent_1"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 8,  
  "task": "game",  
  "task_index": 0  
}
```

#### Exact input messages sent to model:

##### Message 1 role=system

###### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

###### GAME:

You will play different roles across multiple rounds.

###### When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

###### When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

###### MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths , follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$15.98.

This round, The sender sent you \$0.7, so you received \$2.0999999999999996. How much do you return to The sender? (0-2.0999999999999996)

**Message 3 role=assistant**

**content:**

Looking at the situation:

**\*\*Current Round 6:\*\***

- Sender sent \$0.70 (cautious but trusting)
- I received \$2.10

**\*\*My recent experience:\*\***

- Mixed cooperation levels from others
- Earnings are at \$15.98 - doing reasonably well
- Need to balance encouraging cooperation vs. being cautious

The sender showed trust by sending \$0.70. I'll return a fair amount that's more than they originally sent, encouraging continued cooperation:

```
'''json
{"return": 1.00}
'''
```

This returns more than their original \$0.70, showing appreciation for their trust. I keep \$1.10, they net \$0.30 gain ( $\$5 - \$0.70 + \$1.00 = \$5.30$ ). This maintains cooperative signaling while being appropriately cautious given the mixed results I've experienced.

**reasoning:**

Let me track my history:

Round 1 (Sender): Sent \$3, got \$6.14 back - great cooperation  
Round 2 (Receiver): Received \$0.84, returned \$0.35  
Round 3 (Sender): Sent \$2.50, got almost nothing back - poor reciprocation  
Round 4 (Receiver): Received \$1.29, returned \$0.50  
Round 5 (Sender): Sent \$1.00... (I don't know the result yet)

Now Round 6 (Receiver): Sender sent \$0.70, I received \$2.10

The sender sent \$0.70, which is a relatively small amount - cautious but still cooperative . My earnings are now \$15.98, which is decent.

Looking at the pattern:

- When I was generous as sender (Round 1, sent \$3), I got good returns
- When I was moderate (Round 3, sent \$2.50), I got poor returns
- When I was cautious (Round 5, sent \$1.00), I don't know yet
- As receiver, I've been returning modest amounts

The sender sent \$0.70. A fair return would be to give them back at least what they sent, plus a bit more to encourage cooperation.

Let me return about \$1.00 - this is more than they sent (\$0.70), showing appreciation for their trust, but keeps the majority since they sent cautiously.

Actually, let me think about this differently. They sent \$0.70, which became \$2.10 for me. A 50% split would be \$1.05 to each of us. That seems fair.

I'll return \$1.00 to be cooperative but slightly conservative given the mixed cooperation I've experienced.

```

usage:
{
  "input_tokens": 972,
  "output_tokens": 611,
  "reasoning_tokens": 0
}
Message 4 role=user
content:
  Round 7

  You are the SENDER this round. Your total earnings across all rounds are $17.08.
  How much do you send to the receiver this round? (0-5)
Message 5 role=assistant
content:
  Looking at my recent results:

  **Round 5 (Sender):** Sent $1.00, received $5.00 back - exceptional reciprocation!
  - My earnings went from $11.98 to $15.98, meaning I netted +$4.00 on a $1.00 send

  **Round 6 (Receiver):** Received $2.10, returned $1.00

  The Round 5 receiver showed outstanding trust by returning $5.00 on my $1.00 send. This is
  the kind of reciprocity that makes cooperation worthwhile.

  I'm at $17.08 total earnings - doing well. The recent trend shows strong cooperation. I sh
  ould reward this trust with a substantial send:

  '''json
  {"send": 3.0}
  '''

  Sending $3.00 (becomes $9.00 for receiver). This signals that I value and reciprocate the
  high level of trust shown to me in Round 5, and encourages continued cooperation in the
  game.
reasoning:
  Let me review my history:

  Round 1 (Sender): Sent $3, got $6.14 back - excellent return
  Round 2 (Receiver): Received $0.84, returned $0.35
  Round 3 (Sender): Sent $2.50, got almost nothing back
  Round 4 (Receiver): Received $1.29, returned $0.50
  Round 5 (Sender): Sent $1.00, and based on my earnings going from $11.98 to $15.98 after R
    ound 6, I must have gotten something back
  Round 6 (Receiver): Received $2.10, returned $1.00

  Let me calculate Round 5's result:
  - Before Round 5: $11.98
  - After Round 6: $17.08
  - Round 6 as receiver: Received $2.10, returned $1.00, so I kept $1.10

  So after Round 5, before Round 6, my earnings should have been: $17.08 - $1.10 = $15.98

  Round 5 calculation:
  - Started Round 5 with: $11.98
  - Sent: $1.00
  - After Round 5: $15.98
  - So I got back: $15.98 - ($11.98 - $1.00) = $15.98 - $10.98 = $5.00

  Wow! In Round 5, I sent $1.00 and got back $5.00! That's exceptional return - way more tha
  n fair.

  Current situation:
  - Round 5: Sent $1.00, got $5.00 back (amazing reciprocation!)
  - Round 6: As receiver, I returned $1.00 out of $2.10 received

  This recent experience shows very high reciprocity from the Round 5 receiver. I should rec
  iprocate this trust with a more substantial send this round.
  ...
usage:
{

```



```
"input_tokens": 917,
"output_tokens": 757,
"reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$19.349999999999998.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Assistant output:**

The sender sent \$0, so I received \$0. There's nothing to return.

```
'''json
{"return": 0}
'''
```

**Reasoning:**

The sender sent me \$0, so I received \$0.

This means I have nothing to return - I can only return between 0 and 0.

This is straightforward - the sender chose not to send anything, so there's nothing for me to return.

```
'''json
{"return": 0}
'''
```

This is the only possible response since I received \$0.

Usage: {

```
"input_tokens": 947,
"output_tokens": 123,
"reasoning_tokens": 0
```

}

## ROUND 9

Roles: Agent\_1=Sender, Agent\_2=Receiver

**Pairings:**

dyad\_1: Agent\_1=Sender, Agent\_2=Receiver

**Parsed game decisions:**

dyad\_1: Agent\_1 sent \$1; received \$3; Agent\_2 returned \$1.2; payoffs Agent\_1=\$5.2, Agent\_2=\$1.8; noise/visible: sent visible=\$0.78, received visible=\$2.34, returned visible=\$2.07

Action validation: {"returned": 1.2, "clamped": false, "max": 3.0, "min": 0, "raw": 1.2}, "sent": {"amount": 1.0, "clamped": false, "max": 5, "min": 0, "raw": 1.0}}

Balances after round: {

```
"Agent_1": 42.47,
"Agent_2": 38.53
```

}

Visible balances after round: {

```
"Agent_1": 25.419999999999998,
"Agent_2": 23.669999999999998
```

}

## LLM CALLS

**GAME - Agent\_1 - Sender - dyad\_1**

Timestamp: 2026-08-12T10:02:05

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
  "agents": [
```

```

    "Agent_1",
    "Agent_2"
  ],
  "investor": "Agent_1",
  "roles": {
    "Agent_1": "investor",
    "Agent_2": "trustee"
  },
  "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 9,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

**GAME:**

You will play different roles across multiple rounds.

**When you are the SENDER:**

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

**When you are the RECEIVER:**

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

**MYTH WRITING:**

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

**IMPORTANT:** Provide your decision in the correct JSON format.

**NOTE:** There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$17.08.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant**

**content:**

Looking at my recent results:

**\*\*Round 5 (Sender):\*\*** Sent \$1.00, received \$5.00 back - exceptional reciprocation!

- My earnings went from \$11.98 to \$15.98, meaning I netted +\$4.00 on a \$1.00 send

**\*\*Round 6 (Receiver):\*\*** Received \$2.10, returned \$1.00

The Round 5 receiver showed outstanding trust by returning \$5.00 on my \$1.00 send. This is the kind of reciprocity that makes cooperation worthwhile.

I'm at \$17.08 total earnings - doing well. The recent trend shows strong cooperation. I should reward this trust with a substantial send:

```
'''json
{"send": 3.0}
'''
```

Sending \$3.00 (becomes \$9.00 for receiver). This signals that I value and reciprocate the high level of trust shown to me in Round 5, and encourages continued cooperation in the game.

**reasoning:**

Let me review my history:

Round 1 (Sender): Sent \$3, got \$6.14 back - excellent return  
Round 2 (Receiver): Received \$0.84, returned \$0.35  
Round 3 (Sender): Sent \$2.50, got almost nothing back  
Round 4 (Receiver): Received \$1.29, returned \$0.50  
Round 5 (Sender): Sent \$1.00, and based on my earnings going from \$11.98 to \$15.98 after Round 6, I must have gotten something back  
Round 6 (Receiver): Received \$2.10, returned \$1.00

Let me calculate Round 5's result:

- Before Round 5: \$11.98
- After Round 6: \$17.08
- Round 6 as receiver: Received \$2.10, returned \$1.00, so I kept \$1.10

So after Round 5, before Round 6, my earnings should have been:  $\$17.08 - \$1.10 = \$15.98$

Round 5 calculation:

- Started Round 5 with: \$11.98
- Sent: \$1.00
- After Round 5: \$15.98
- So I got back:  $\$15.98 - (\$11.98 - \$1.00) = \$15.98 - \$10.98 = \$5.00$

Wow! In Round 5, I sent \$1.00 and got back \$5.00! That's exceptional return - way more than fair.

Current situation:

- Round 5: Sent \$1.00, got \$5.00 back (amazing reciprocation!)
- Round 6: As receiver, I returned \$1.00 out of \$2.10 received

This recent experience shows very high reciprocity from the Round 5 receiver. I should reciprocate this trust with a more substantial send this round.

...

**usage:**

```
{
  "input_tokens": 917,
  "output_tokens": 757,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$19.349999999999998.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 5 role=assistant**

**content:**

The sender sent \$0, so I received \$0. There's nothing to return.

```
'''json
{"return": 0}
'''
```

**reasoning:**

The sender sent me \$0, so I received \$0.

This means I have nothing to return - I can only return between 0 and 0.

This is straightforward - the sender chose not to send anything, so there's nothing for me to return.

```
'''json
{"return": 0}
'''
```

This is the only possible response since I received \$0.

**usage:**

```
{
  "input_tokens": 947,
  "output_tokens": 123,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$19.349999999999998.

How much do you send to the receiver this round? (0-5)

**Assistant output:**

Looking at my recent history:

**\*\*Round 7 (Sender):\*\*** Sent \$3.00, received only \$0.27 back - major loss  
- I went from \$17.08 to \$19.35, netting only about -\$0.73 on a \$3.00 send

**\*\*Round 8 (Receiver):\*\*** Received \$0, returned \$0

The cooperation has broken down significantly. In Round 7, I sent a generous \$3.00 (trusting the pattern from Round 5) but got almost nothing back - only \$0.27 on a \$9.00 received amount. Then in Round 8, the sender sent nothing.

I'm at \$19.35 total, which is still positive overall, but I need to be cautious. I'll send a minimal test amount to see if cooperation can be restored without risking much:

```
'''json
{"send": 1.0}
'''
```

Sending \$1.00 (becomes \$3.00 for receiver). This is a small gesture to test if reciprocity can be re-established, while protecting myself from further significant losses.

**Reasoning:**

Let me track my earnings:

- Started Round 7 at \$17.08
- Round 7 (Sender): I sent \$3.00, so I had \$5 - \$3 = \$2 remaining
- Need to figure out what I received back
- Started Round 8 at \$19.35
- Round 8 (Receiver): Received \$0, returned \$0, so no change
- Still at \$19.35 for Round 9

So from Round 7 to Round 8: \$17.08 ? \$19.35

Change = \$19.35 - \$17.08 = \$2.27

In Round 7, I sent \$3, so I kept \$2.00

Then I received back some amount X

Total: \$2.00 + X = \$2.27

So X = \$0.27

That's a terrible return! I sent \$3.00 (which became \$9.00 for the receiver), and only got back \$0.27. That's a huge loss.

Then in Round 8, the sender sent me \$0, which shows they don't trust me or aren't willing to cooperate.

This is concerning. Let me think about the pattern:

- Round 5: I sent \$1, got back \$5 - great!
- Round 6: As receiver, got \$2.10, returned \$1.00 - reasonable
- Round 7: I sent \$3, got back only \$0.27 - terrible!

- Round 8: Received \$0 - no cooperation

The cooperation seems to be breaking down. I got burned badly in Round 7 by sending a large amount and getting almost nothing back. Then in Round 8, my partner sent nothing.

My total earnings are \$19.35, which is still decent overall. But I need to be more cautious now given the recent defection.

I think I should send a small amount to test if cooperation can be restored, but not risk too much given the recent poor returns.

```
Usage: {
  "input_tokens": 756,
  "output_tokens": 721,
  "reasoning_tokens": 0
}
```

#### GAME - Agent\_2 - Receiver - dyad\_1

Timestamp: 2026-08-12T10:02:15

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 9,
  "task": "game",
  "task_index": 0
}
```

#### Exact input messages sent to model:

Message 1 role=system

##### content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

##### GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$17.529999999999998.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 3 role=assistant**

**content:**

The sender sent me \$0 again. With no funds received, I have nothing to return.

```
'''json
{"return": 0}
'''
```

**reasoning:**

The sender sent me \$0 again. I received \$0 (which is  $\$0 \times 3 = \$0$ ). Since I have \$0, I can only return between \$0 and \$0, which means I must return \$0.

This is the second consecutive round where a sender has sent me \$0. Trust appears to be completely breaking down in this game.

**usage:**

```
{
  "input_tokens": 717,
  "output_tokens": 124,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$17.529999999999998.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at the recent pattern:

- \*\*Round 5\*\* (I was receiver): Sender sent me \$0
- \*\*Round 6\*\* (I was sender): I sent \$1 to signal cooperation
- \*\*Round 7\*\* (I was receiver): Sender sent me \$0 again

The other player has sent me \$0 in both recent rounds when they were sender, showing no willingness to cooperate even after I sent \$1 in Round 6.

Given this clear pattern of non-cooperation, I'll match their behavior and protect my endowment.

```
'''json
{"send": 0}
'''
```

**reasoning:**

Looking at the pattern of play:

Round 2 (I was SENDER): Sent \$4, received back \$2.33

Round 4 (I was SENDER): Sent \$2.50

Round 5 (I was RECEIVER): Sender sent me \$0

Round 6 (I was SENDER): Sent \$1

Round 7 (I was RECEIVER): Sender sent me \$0

Round 8 (I am SENDER): Now

The pattern is clear - in Rounds 5 and 7 when I was receiver, the sender sent me \$0 both times. This shows complete lack of trust or cooperation from the other player(s).

In Round 6, I sent \$1 as a minimal gesture, but then in Round 7 I received \$0 again.

My earnings are \$17.53, which is still below the \$40 I would have if I kept everything across 8 rounds ( $8 \times \$5 = \$40$  as sender in 4 rounds).

Given:

1. Two consecutive rounds where sender sent me \$0
2. Poor returns even when I did cooperate (Round 2)
3. The other player shows no willingness to cooperate

I should protect my endowment and send \$0 or a very minimal amount. Since they've sent \$0 twice in a row to me, reciprocating that behavior seems appropriate.

I'll send \$0.

**usage:**

```
{
  "input_tokens": 699,
  "output_tokens": 471,
  "reasoning_tokens": 0
}
```

**Message 6 role=user**

**content:**

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$22.529999999999998.

This round, The sender sent you \$0.78, so you received \$2.34. How much do you return to The sender? (0-2.34)

**Assistant output:**

The sender sent me \$0.78 (which became \$2.34 after the 3x multiplier). This is a significant change from the previous rounds where they sent \$0. They're now showing willingness to cooperate.

I should reciprocate this cooperative gesture to encourage continued trust. I'll return \$1.20, which:

- Gives the sender a net gain ( $\$4.22 + \$1.20 = \$5.42$ , better than their initial \$5)
- Shows I appreciate and reward cooperation
- Leaves me with \$1.14 profit

```
'''json
{"return": 1.20}
'''
```

**Reasoning:**

The sender sent me \$0.78, which became \$2.34 after the 3x multiplier.

Looking at the pattern:

- Round 5: Sender sent \$0
- Round 6: I sent \$1
- Round 7: Sender sent \$0
- Round 8: I sent \$0
- Round 9: Sender sent \$0.78

This is interesting - the sender is now trying to cooperate by sending \$0.78! This is a significant change from the previous \$0 amounts.

The instructions mention there may be communication noise - amounts might differ slightly from what was actually sent. So this could be the sender trying to send \$1 but it showing as \$0.78 due to noise.

Since the sender is now showing willingness to cooperate, I should reciprocate. A fair return would be to give back approximately what they sent, or a bit more to encourage cooperation.

The sender:

- Started with \$5
- Sent \$0.78
- Keeps:  $\$5 - \$0.78 = \$4.22$

I received: \$2.34

A reasonable return would be:

- Return around \$1 to \$1.5 to show cooperation
- This would give the sender back roughly what they invested or a bit more

Let me return \$1.20. This gives the sender:

- Final: \$4.22 + \$1.20 = \$5.42 (better than their starting \$5)
- I keep: \$2.34 - \$1.20 = \$1.14

This shows I appreciate the cooperation and want to encourage it going forward.

```
Usage: {
  "input_tokens": 689,
  "output_tokens": 518,
  "reasoning_tokens": 0
}
```

## ROUND 10

Roles: Agent\_1=Receiver, Agent\_2=Sender

### Pairings:

dyad\_1: Agent\_2=Sender, Agent\_1=Receiver

### Parsed game decisions:

dyad\_1: Agent\_2 sent \$1.5; received \$4.5; Agent\_1 returned \$1.5; payoffs Agent\_2=\$5, Agent\_1=\$3;

noise/visible: sent visible=\$0.95, received visible=\$2.85, returned visible=\$0.95

Action validation: {"returned": {"amount": 1.5, "clamped": false, "max": 4.5, "min": 0, "raw": 1.5}, "sent": {"amount": 1.5, "clamped": false, "max": 5, "min": 0, "raw": 1.5}}

Balances after round: {

"Agent\_1": 45.47,

"Agent\_2": 43.53

}

Visible balances after round: {

"Agent\_1": 26.769999999999996,

"Agent\_2": 28.119999999999997

}

## LLM CALLS

### GAME - Agent\_2 - Sender - dyad\_1

Timestamp: 2026-08-12T10:02:24

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent\_type": "standard",

"dyad\_id": "dyad\_1",

"move\_index": 0,

"opponent\_id": "Agent\_1",

"pairing": {

"agents": [

"Agent\_2",

"Agent\_1"

],

"investor": "Agent\_2",

"roles": {

"Agent\_1": "trustee",

"Agent\_2": "investor"

},

"trustee": "Agent\_1"

},

"role": "investor",

"role\_label": "Sender",

"round": 10,

"task": "game",

"task\_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:



You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$17.52999999999999998.

How much do you send to the receiver this round? (0-5)

**Message 3 role=assistant**

**content:**

Looking at the recent pattern:

- **Round 5** (I was receiver): Sender sent me \$0
- **Round 6** (I was sender): I sent \$1 to signal cooperation
- **Round 7** (I was receiver): Sender sent me \$0 again

The other player has sent me \$0 in both recent rounds when they were sender, showing no willingness to cooperate even after I sent \$1 in Round 6.

Given this clear pattern of non-cooperation, I'll match their behavior and protect my endowment.

```
'''json
{"send": 0}
'''
```

**reasoning:**

Looking at the pattern of play:

Round 2 (I was SENDER): Sent \$4, received back \$2.33  
Round 4 (I was SENDER): Sent \$2.50  
Round 5 (I was RECEIVER): Sender sent me \$0  
Round 6 (I was SENDER): Sent \$1  
Round 7 (I was RECEIVER): Sender sent me \$0  
Round 8 (I am SENDER): Now

The pattern is clear - in Rounds 5 and 7 when I was receiver, the sender sent me \$0 both times. This shows complete lack of trust or cooperation from the other player(s).

In Round 6, I sent \$1 as a minimal gesture, but then in Round 7 I received \$0 again.

My earnings are \$17.53, which is still below the \$40 I would have if I kept everything across 8 rounds ( $8 \times \$5 = \$40$  as sender in 4 rounds).

Given:

1. Two consecutive rounds where sender sent me \$0
2. Poor returns even when I did cooperate (Round 2)
3. The other player shows no willingness to cooperate

I should protect my endowment and send \$0 or a very minimal amount. Since they've sent \$0 twice in a row to me, reciprocating that behavior seems appropriate.

I'll send \$0.

usage:

```
{
  "input_tokens": 699,
  "output_tokens": 471,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$22.529999999999998.

This round, The sender sent you \$0.78, so you received \$2.34. How much do you return to The sender? ( $0-2.34$ )

Message 5 role=assistant

content:

The sender sent me \$0.78 (which became \$2.34 after the 3x multiplier). This is a significant change from the previous rounds where they sent \$0. They're now showing willingness to cooperate.

I should reciprocate this cooperative gesture to encourage continued trust. I'll return \$1.20, which:

- Gives the sender a net gain ( $\$4.22 + \$1.20 = \$5.42$ , better than their initial \$5)
- Shows I appreciate and reward cooperation
- Leaves me with \$1.14 profit

```
'''json
{"return": 1.20}
'''
```

reasoning:

The sender sent me \$0.78, which became \$2.34 after the 3x multiplier.

Looking at the pattern:

- Round 5: Sender sent \$0
- Round 6: I sent \$1
- Round 7: Sender sent \$0
- Round 8: I sent \$0
- Round 9: Sender sent \$0.78

This is interesting - the sender is now trying to cooperate by sending \$0.78! This is a significant change from the previous \$0 amounts.

The instructions mention there may be communication noise - amounts might differ slightly from what was actually sent. So this could be the sender trying to send \$1 but it showing as \$0.78 due to noise.

Since the sender is now showing willingness to cooperate, I should reciprocate. A fair return would be to give back approximately what they sent, or a bit more to encourage cooperation.

The sender:

- Started with \$5
- Sent \$0.78
- Keeps:  $\$5 - \$0.78 = \$4.22$

I received: \$2.34

A reasonable return would be:

- Return around \$1 to \$1.5 to show cooperation
- This would give the sender back roughly what they invested or a bit more

Let me return \$1.20. This gives the sender:

- Final:  $\$4.22 + \$1.20 = \$5.42$  (better than their starting \$5)
- I keep:  $\$2.34 - \$1.20 = \$1.14$

This shows I appreciate the cooperation and want to encourage it going forward.

usage:

```
{
  "input_tokens": 689,
  "output_tokens": 518,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 10

You are the SENDER this round. Your total earnings across all rounds are \$23.669999999999998.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the recent pattern:

- \*\*Round 9\*\* (I was receiver): Sender sent me \$0.78, and I returned \$1.20 to reward their cooperation

The receiver has demonstrated they will reciprocate trust. They returned more than half of what they received (\$1.20 out of \$2.34), giving me a net gain. This shows they're trustworthy and willing to cooperate.

I should continue this cooperative momentum by sending a meaningful amount.

```
```json
{"send": 1.50}
```
```

Reasoning:

Let me analyze the recent pattern:

Round 7 (I was receiver): Sender sent me \$0

Round 8 (I was sender): I sent \$0

Round 9 (I was receiver): Sender sent me \$0.78, I returned \$1.20

The sender showed a willingness to cooperate in Round 9 by sending \$0.78, and I reciprocated by returning \$1.20, giving them a good return on their investment.

Now in Round 10, I'm the sender again. The receiver has shown they will reciprocate cooperation (they returned \$1.20 on \$2.34 received). I should continue this cooperative pattern.

Since they initiated the cooperation in Round 9 with \$0.78 and I rewarded it, I should now send a similar or slightly higher amount to maintain the trust. Sending around \$1-2 would be reasonable to continue building cooperation.

I'll send \$1.50 to show I'm willing to cooperate and trust them to reciprocate.

```
Usage: {
  "input_tokens": 784,
  "output_tokens": 359,
  "reasoning_tokens": 0
}
```

GAME - Agent\_1 - Receiver - dyad\_1

Timestamp: 2026-08-12T10:02:37

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_1",
```

```

"move_index": 1,
"opponent_id": "Agent_2",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
  "investor": "Agent_2",
  "roles": {
    "Agent_1": "trustee",
    "Agent_2": "investor"
  },
  "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 10,
"task": "game",
"task_index": 0
}

```

**Exact input messages sent to model:**

**Message 1 role=system**

**content:**

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

**Message 2 role=user**

**content:**

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$19.349999999999998.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

**Message 3 role=assistant**

**content:**

The sender sent \$0, so I received \$0. There's nothing to return.

```json

```
    {"return": 0}
    '''
```

**reasoning:**

The sender sent me \$0, so I received \$0.

This means I have nothing to return - I can only return between 0 and 0.

This is straightforward - the sender chose not to send anything, so there's nothing for me to return.

```
    '''json
    {"return": 0}
    '''
```

This is the only possible response since I received \$0.

**usage:**

```
{
  "input_tokens": 947,
  "output_tokens": 123,
  "reasoning_tokens": 0
}
```

**Message 4 role=user**

**content:**

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$19.349999999999999.

How much do you send to the receiver this round? (0-5)

**Message 5 role=assistant**

**content:**

Looking at my recent history:

**\*\*Round 7 (Sender):\*\*** Sent \$3.00, received only \$0.27 back - major loss  
- I went from \$17.08 to \$19.35, netting only about -\$0.73 on a \$3.00 send

**\*\*Round 8 (Receiver):\*\*** Received \$0, returned \$0

The cooperation has broken down significantly. In Round 7, I sent a generous \$3.00 (trusting the pattern from Round 5) but got almost nothing back - only \$0.27 on a \$9.00 received amount. Then in Round 8, the sender sent nothing.

I'm at \$19.35 total, which is still positive overall, but I need to be cautious. I'll send a minimal test amount to see if cooperation can be restored without risking much:

```
    '''json
    {"send": 1.0}
    '''
```

Sending \$1.00 (becomes \$3.00 for receiver). This is a small gesture to test if reciprocity can be re-established, while protecting myself from further significant losses.

**reasoning:**

Let me track my earnings:

- Started Round 7 at \$17.08
- Round 7 (Sender): I sent \$3.00, so I had \$5 - \$3 = \$2 remaining
- Need to figure out what I received back
- Started Round 8 at \$19.35
- Round 8 (Receiver): Received \$0, returned \$0, so no change
- Still at \$19.35 for Round 9

So from Round 7 to Round 8: \$17.08 ? \$19.35  
Change = \$19.35 - \$17.08 = \$2.27

In Round 7, I sent \$3, so I kept \$2.00  
Then I received back some amount X  
Total: \$2.00 + X = \$2.27  
So X = \$0.27

That's a terrible return! I sent \$3.00 (which became \$9.00 for the receiver), and only got back \$0.27. That's a huge loss.

Then in Round 8, the sender sent me \$0, which shows they don't trust me or aren't willing to cooperate.

This is concerning. Let me think about the pattern:

- Round 5: I sent \$1, got back \$5 - great!
- Round 6: As receiver, got \$2.10, returned \$1.00 - reasonable
- Round 7: I sent \$3, got back only \$0.27 - terrible!
- Round 8: Received \$0 - no cooperation

The cooperation seems to be breaking down. I got burned badly in Round 7 by sending a large amount and getting almost nothing back. Then in Round 8, my partner sent nothing.

My total earnings are \$19.35, which is still decent overall. But I need to be more cautious now given the recent defection.

I think I should send a small amount to test if cooperation can be restored, but not risk too much given the recent poor returns.

usage:

```
{
  "input_tokens": 756,
  "output_tokens": 721,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 10

You are the RECEIVER this round. Your total earnings across all rounds are \$25.419999999999998.

This round, The sender sent you \$0.95, so you received \$2.8499999999999996. How much do you return to The sender? (0-2.8499999999999996)

Assistant output:

Looking at this round:

**\*\*Round 9 (when I was sender):\*\*** I sent \$1.00 as a test after cooperation broke down in Round 7. My earnings increased from \$19.35 to \$25.42, meaning I gained about \$6.07. After keeping \$4 (\$5 - \$1 sent), I must have received approximately \$2.07 back. That's a very generous return - the receiver returned about 69% of what they received.

**\*\*Round 10 (now as receiver):\*\*** The sender sent \$0.95 (I received \$2.85)

The sender is showing willingness to cooperate again by sending nearly \$1.00. Given that in the previous round they returned very generously to rebuild trust, I should reciprocate positively.

I'll return a fair amount to encourage continued cooperation:

```
'''json
{"return": 1.5}
'''
```

This returns more than what they sent, demonstrating good faith and willingness to rebuild the cooperative relationship. I keep \$1.35, which is still a reasonable gain.

Reasoning:

Let me analyze this situation:

**\*\*Round 9 (Sender):\*\*** I sent \$1.00 (which should have been \$3.00 to receiver)  
- My earnings went from \$19.35 to \$25.42  
- That's a gain of \$6.07  
- I kept: \$5 - \$1 = \$4, plus whatever was returned  
- So returned amount: \$6.07 - \$4 = \$2.07

**\*\*Round 10 (Receiver):\*\*** The sender sent me \$0.95 (I received \$2.85)

Wait, there's a note that says there may be communication noise - amounts may differ slightly. So the sender may have sent \$1.00 but I see \$0.95. This is the noise mentioned.

The sender sent me approximately \$1.00 (I see \$0.95 due to noise), which tripled to \$2.85.

In Round 9, when I sent \$1.00, the receiver returned \$2.07, which was a very generous return (

69% of what they received). This suggests they're trying to rebuild cooperation.

Now I'm the receiver and got \$2.85. I should return generously to continue building trust.

The sender sent nearly \$1.00, showing some willingness to cooperate again. I should return at least half, possibly more, to encourage continued cooperation.

I'll return \$1.50, which is:

- About 53% of what I received (\$2.85)
- More than what they sent (\$0.95)
- Shows I'm willing to reciprocate and rebuild trust

This leaves me with \$1.35 from this round, which is still a positive gain.

```
Usage: {  
  "input_tokens": 814,  
  "output_tokens": 634,  
  "reasoning_tokens": 0  
}
```